N E W S

GREENSTAR NATIONAL COMPUTER CLUB WISH TO THANK
BRENDA SOLOMON AND STAFF OF
JAMES RALPH (PTY) LTD.,
THE IMPORTERS AND DISTRIBUTORS OF SPECTRAVIDEO AND
BONDWELL COMPUTERS FOR THEIR VALUABLE SUPPORT.
WITHOUT THIS SUPPORT, THIS CLUB WOULD <NEVER>
HAVE COME INTO BEING.

WE WISH EVERYONE AT JAMES RALPH THE VERY BEST FOR THE FUTURE.

****

GREENSTAR WISH TO ANNOUNCE THAT DUE TO THE PROPOSED REVISION OF THE EDUCATIONAL SYSTEM IN S.A., IT HAS BEEN DECIDED THAT, ALL SCHOOL-ORIENTATED PROGRAMS WILL BE TEMPORARILY WITHDRAWN FROM THE SOFTWARE LIST.
ENQUIRIES IN THIS RESPECT ARE STILL WELCOME VIA THE HOTLINE.

****

DUE TO LACK OF SPACE THIS MONTH, THE USUAL MONTHLY PROJECT WILL NOT APPEAR. THE PROJECT WILL HOWEVER, APPEAR IN NEXT MONTH'S NEWSLETTER.

****

B A C K - I S S U E S

BACK-ISSUES OF THE NEWSLETTER FROM JANUARY '86 ARE STILL AVAILABLE AT R2.00 PER COPY (incl. GST AND POSTAGE).

****

NEWS OF SPECTRAVIDEO'S MSX 3.0 TO BE RELEASED SHORTLY.

ACCORDING TO EUROPEAN COMPUTER MAGAZINE, SPECTRAVIDEO INTERNATIONAL IS PRODUCING AN MSX 3.0 COMPUTER WHICH, IT APPEARS, WILL BE MS-DOS AND MSX COMPATIBLE.
IT WILL HAVE MORE MEMORY THAN MSX 2.0 AND HAVE A 16-BIT CO-PROCESSOR SUCH AS INTEL-8088 OR BETTER.
WE HAVE NOT BEEN ABLE TO ASCERTAIN WHETHER THE 3-1/2 INCH DRIVE WILL BE BUILT IN OR SUPPLIED AS A SEPARATE UNIT. IF THE X'PRESS IS ANYTHING TO GO BY, THIS WILL BE A COMPUTER, SECOND-TO-NONE
MS-DOS, MSX-DOS, CP/M, BASIC.....WOW !!!!!

****

EVERY man is a story which is identical with no other. – Alexis Carrel

A REAL test of will power is to have the same ailment someone is describing—and not mention it. – Bob Crowe in *Star Free Press*

GREENSTAR NATIONAL COMPUTER CLUB

(BASED IN THE VAAL-TRIANGLE)

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CLUB MONTHLY NEWSLETTER

30th AUG.'86

SOFTWARE LIBRARY

MSX

SVI-328

Software Bonanza

SD233C/T	Armoured Assault
SD231T	Artist
SD305T	Boa
SD307T	CarAce
SD291C	Flipper Slipper
SD232C	Frantic Freddy
SD310T	Horse Races
SD306T	Juno Lander
SD293T	Kung Fu Master
SD236C	Music Mentor
SD219T	New York Bomb Blitz
SD292T	Ninja
SD303T	Number Game
SD211T	Old Mac Farmer
SD243T	Sasa
SD220C	Sector Alpha
SD309T	Spectra Break
SD234T	Spectron
SD237C	Super Cross Force
SD241T	Telebunny
SD216T	Tetra Horror
SD242T	Turboat

creative computing®

the #1 magazine of computer applications and software

Test: Spectravideo SV-318 and SV-328

The Spectravideo family of computers is an interesting international effort. Overall management, marketing, and direction come from the company head-quarters in New York; systems software development including the powerful implementation of Microsoft Basic was carried out by ASCII Microsoft in Tokyo; while the manufacturing is done in Hong Kong. The SV-318 and SV-328 are the products of this unusual, but highly successful three-way marriage.

While this is a review of both machines, we will generally speak of the computer (singular) and point out the difference between the two models (built-in memory and keyboard) in the appropriate places.

Functional Console

The computer is housed in a sturdy white plastic case measuring 15.5" × 8.7" which slopes from a height of 1.7" at the front to 3" at the rear. Immediately visible on the top of the case is the keyboard and a red LED indicating that the power is on. Although a small feature, this is much appreciated; so many computers today provide no indication of whether they are on or off.

also on top of the case is a hinged cartridge port in which to plug in software packages that come in cartridge form.

Moving to the right side, we find a male 4-prong power supply input. The power supply is a heavy duty unit providing 14 volts at 2 amps and 8 volts at 3 amps. This is three to four times the power offered by the average home computer power supply; thus it ought to more than enough reserve power for all the peripherals you may ever want to add.

Also on the right side are a rocker-type off-on power switch and two game controller ports. These are standard DB-9 receptacles for A turn-type joysticks, paddle controls, and track balls.

On the bank of the console are an expander port, cassette port and video output port. An RF modulator furnished with the system plugs into the video port and, according to the specs, will produce NTSC, PAL or SECAM output as appropriate. It uses a standard 5-pin, 180-degree DIN connector which can be easily wired to provide a composite video signal to a monitor with separate audio output.

The cassette port connects to the Spectravideo stereo data/audio cassette recorder. The computer has lines for sending and receiving data, motor control, and power.

The expander port allows the computer to be connected to a single slot expansion adapter or a 7-slot expansion box. More about this later.

A Complete Package

The attractive box contains nearly everything you need to start computing. You get the computer/keyboard unit, RF modulator (with 3" cable), shielded video cable (5"), video switch box, power supply, instruction manual, and warranty card.

The manual has eight pages of photos depicting all aspects of hooking up the system and getting it into operation. Another helpful feature is the built-in diagnostic check that automatically occurs upon powering up the system. A successful self-test is signaled by a beep in the TV speaker about 1-1/2 seconds after turning on the power switch.

Upon a successful power up, the Spectravideo logo appears in three colors followed by a message indicating the version of Microsoft Basic in the system and the number of free bytes of user memory. A standard SV-318 has 32K of RAM; 16K is allocated to graphics support and the other 16K is user addressable memory. Well, not quite, 3569 bytes are reserved for overhead, I/O, and the like. Thus an unexpanded 16K SV-318 has about 12.8K truly usable bytes and a standard, 32K SV-328 machine has 29.2K. Both computers are expandable, with 16K or 64K external memory packs, to a maximum of 144K.

Computer/Keyboard Unit

The SV computers use the Z80A mpu chip operating at 3.6 MHz; until recently, most peronal computers used a 2 MHz clock rate. In theory, this should make the SV computers faster than other comparable machines. In practice, the issue of speed is somewhat more complicated.

The version of Microsoft Basic used by the SV computers automatically makes all variables double precision. Hence, a variable occupies eight bytes, and every time it is called, those eight bytes must be moved. In contrast, most other small computers use single precision variables (four bytes) so there are correspondingly fewer memory accesses when a variable is called. The bottom line is that in a normal Basic program, the SV computers are considerably more accurate than, say, the Mattel Aquarius or TRS-80 Model 4, but only about half as fast. A slight improvement in speed (about 10%) can be obtained by defining variables as single precision.

On the other hand, the speed advantage is apparent when using machine code. Also, as we will see later, the inclusion of several powerful Basic and graphics commands makes programming much simpler, particularly for animated graphics. This may yield a speed advantage as well, since one command does the job of many.

The keyboard on the SV-318 has 71 rubberized Chiclet-style keys. in addition to the expected alphabetic , numeric, and symbol keys, the keyboard has five function keys, three program control keys (STOP, ENTER, and CONTROL), and six miscellaneous keys.

Each of the five control keys activates two functions depending upon whether SHIFT Is pressed. The function of each of these keys is shown on the bottom of the TV display. The functions controlled by these keys are color (to set character, border, and background colors), auto line numbering, list (the entire program or the last line you were working on), run, cassette load, GOTO (allows execution of a program from any point), and continue. In addition, all ten of the function keys are user programmable with a simple statement, for example, KEY 1, "Creative". From then on, whenever function key 1 is pressed, the word Creative will automatically appear.

The six miscellaneous keys are caps lock, clear screen and move cursor to home position, insert, delete, and left and right graphics. These last two keys are used to select the 52 graphic symbols on the keyboard (each of the 26 letters can produce two graphics symbols).

To the right of the keyboard on the SV-318 is a joystick/cursor control pad. A joystick may be screwed into the center of the pad for use in games or to move the cursor around the screen in any of eight directions. Actually, we found it most convenient to move the cursor by pressing the indentations in the pad rather than using the joystick at all.

The feel of the keyboard was as good as could be expected from the Chiclet-style keys. With each keypress, an audible keyclick sound is produced in the TV speaker to aid in accurate typing. This can be turned on or off with the commands CLICK ON or CLICK OFF.

The SV-328 differs from the SV-318 in that it has a full-stroke keyboard with 86 keys. In addition to the keys on the SV-318, the SV-328 has a numeric keypad and arithmetic function keys to the right of the main keyboard, and the cursor control pad is replaced by four directional keys. The keyboard has an excellent tactile feel, and we had no need to leave the keyclick sound on.

Extended Microsoft Basic

The version of Microsoft Basic in the SV computers is one of the richest we have ever seen. As mentioned earlier, all variables are automatically double precision unless specified otherwise in your program. This yields 16.8 decimal digits of accuracy which beats hands down any computer that we have tested with the exception of the calculator-like TI CC 40.

The computer has four types of variables: double precision, single precision, integer, and string. A variable type can be declared at the beginning of the program (DEFSNG, DEFINT, etc.) or by means of a suffix attached to the variable name (I%, B/, F#, A$).

There are 26 Basic commands. These include the usual NEW, RUN, SAVE, and the like, but we found a number of unusual commands as well.

KEY LIST lists the contents of the programmable function keys. MERGE brings a second program into memory and merges it with the one already there; if there are duplicate line numbers, the second program takes precedence.

MOTOR ON or OFF turns the cassette motor on or off

SWITCH causes the computer to use a different memory bank. WIDTH sets the width of the screen display and TRON and TROFF turn the trace function on and off.

An extended list of 29 Basic statements includes BEEP and SOUND (R.B) which puts a sound byte, B, into one of the three sound registers, R. Control of the three sound channels is extensive and includes pitch (over eight octaves), amplitude, envelope period, envelope shape, and rhythm. In addition, there is a noise generator. To take full advantage of this sound sound capability, the SV computers include a music Macro Language with 11 additional commands.

Other extended Basic statements include SWAP (exchanges the value of two variables), WAIT (suspends program execution to read an input port), OUT (puts a byte to an output port), DEF USR (defines an entry address for a machine language subroutine), and ERASE (releases space used by a variable array).

The Basic operators include the expected arithmetic and Boolean operations. In addition, we find MOD (integer modulus), XOR (exclusive OR), EQV (equivalence), and IMP (implication). The arithmetic functions are what would be expected; however, Spectravideo has thoughtfully provided in the User's Manual a table of 20 inverse and hyper-bolic functions not directly implemented on the computer, showing the formula to calculate each one.

Again, the string functions are more or less as expected with the addition of HEX$ and OCT$ (converts numbers to hexadecimal and octal strings), and STRINGS$ (L, E) which returns a string of L length with the numeric value specified by expression E.

The I/O and interrupt control functions and commands are fascinating and allow amazing control of printers, terminals, joysticks, and other I/O devices. The SV computers have a built-in timer which can be accessed from Basic programs to perform all kinds of tricks. We were especially interested in the statement ON SPRITE GOSUB which causes the program to jump to an address when it detects a collision between two sprites.

Sprites are little graphics critters which can consist of up to 32 bytes which define

colors and pixels on the screen (rockets, tanks, people, or any moving object). A program can use up to 32 sprites, many more than are available in other comparably priced computers. Unfortunately, neither the User's Guide nor the Quick Reference Guide included with the computer describes how to use sprites in a program.

In addition on the various sprite commands, the SV computers have many other graphics statements which make graphics programming a real joy. These include CIRCLE, LINE, GET, POINT, PSET, VPEEK and VPOKE (peek and poke to video screen locations), and DRAW. This last command is used to draw on the screen with a special graphics macro language which has 14 additional commands.

On-Screen Editing

More and more compters are being produced today with on-screen editing to replace the older systems which required an entire statement to be retyped or a special editing function to be invoked. With on-screen editing, you simply move the cursor to the place requiring a change, type the change, press RETURN, and presto, the change is made. Good on-screen systems include insert and delete keys, and allow the duplication of statements by simply typing a new line number over the old one; the SV computers have these features.

In addition, the SV computers have several other editing and cursor movement functions which are invoked by pressing the CONTROL key in combination with a letter. Some of these functions include backspace-and-delete, cursor to end of line, truncate line (a real joy!), and clear logical line.

The only "missing" editing command that we would like to have seen is Line Insert; however, the only low cost computer on which we have found this command is the Panasonic JR-200.

Graphics Display

As mentioned earlier, the SV computers can drive either a color monitor or a TV set with an RF signal. Naturally the image on a monitor is better, but the computer produces a surprisingly good image on a TV set. Up to 16 colors can be produced simultaneously, although some of them tend to appear very similar. On the other hand, it is rare to want 16 completely different colors on the screen at one time.

The default color for text is a highly legible white on blue, although by means of the color command you can set this to anything you desire.

Normal text resolution is 40 characters by 24 lines. In Basic, the bottom line is reserved for the function key definitions.

There are two graphics modes, low-and high-resolution, appropriately enough. High-resolution provides 256 × 192 pixels; low- resolution has 64 × 48 boxes. In addition, you can use the graphics characters in text mode (40 × 24). While this sounds practically useless, bear in mind that the 52 graphics characters effectively divide each box into four; thus the usable resolution is more like 80 × 48.

Documentation

If there is a weakness in the Spectravideo line, it is in documentation. The computer comes with a 136-page User's Manual and 22-page Quick Reference Guide. Both are punched for a 6" × 9"D three-ring binder (included), but the spiral bound User's Guide is better used without the binder.

The User's Manual takes a very different approach to teaching Basic than we have seen before. In particular, it starts with graphics and shows how you can use the computer to design simple pictures and draw them on the screen. This is an excellent approach and builds up interest quickly among new users.

Unfortunately, it doesn't go far enough. While it deals with all the standard Basic and graphics commands, it doesn't discuss any of the nifty extended features of the SV computers. Nowhere are we told how to use sprites. The amazing sound capabilities of the machine are covered in only the most cursory way.

In Marseilles, France, a fellow who makes his living painting the exterior of shops and stores has been injured three times by falls from his ladder. "People were so busy looking into shop windows," he explains, "that they didn't see my ladder and would accidentally ram shopping trolleys and prams into it."

Eventually the Frenchman resorted to putting this warning sign in front of his ladder: "Danger. Nude Painter Above."

"People certainly notice me now," he reports.

On my first day of work in a chemical-research laboratory, I was intrigued by some complicated apparatus at the end of my workbench. It consisted of some expensive-looking glassware projecting from a five-litre flask that nestled in an electric-heating mantle. Every few minutes the experiment received the attention of a white-coated scientist who made a few adjustments. By midmorning, I could contain my curiosity no longer and asked what the desired result was.

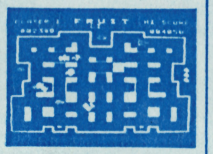
"Coffee," was the reply. "This is our percolator. Would you like a cup?"

– Philip Clarke, Eikenhof, Transvaal

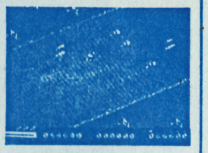
SOFTWARE LIBRARY FOR SVI-328

211T OLD MAC FARMER
Help Old Mac Farmer clear his field of pesty bugs by destroying their eggs with your limited supply of insecticide. Then, protect the crops from the constant attack of giant birds and nasty ducks.

216T TETRA HORROR
Face fierece skulls and ghosts and avoid rolling boulders and deadly pits in your role as TAMO the adventure. Use your skill and fire gun to destroy your enemies, while battling your way through the TETRA CASTLE to obtain treasure.

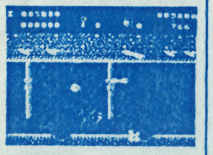


241T TELEBUNNY
Benny bunny guides the hapless turtle through an intricate maze while avoiding snakes and broken hearts in this exciting and challenging arcade game.

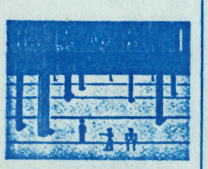


242T TURBOAT
Your Turboat must avoid missles and torpedo in a mission to penetrate enemy forts, destroy jetfoils and pickup secret radars.

243T SASA
You control SASA the robot as he breaks into the enemy arsenal and destroys the power generators while battling gravity, an insatiable thirst for fuel and a variety of other obstacles to survive.



292T NINJA
The ancient Ninja warriors of Japan fight it out in this epic battle for fame and fortune.

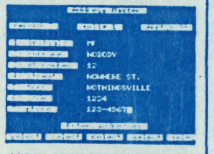


293T KUNG FU MASTER
Exercise your skill in the martial arts. Stave off your enemies, the Dummies, as they attack you from all sides, with Kicks, punches and even flying darts, to become " the Kung fu master".

FIRST STEP SD222T
First step is a fun and easy way to help your child learn letter and number recognition, and early arithmetic skills.

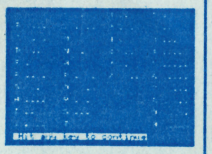
BUSY BEE SD226T
My multi-colored Busy Bee with a keyboard, teach young children to create colorful computer art.

602D SPRITE MASTER
SPRITE MASTER is a handy utility for the would-be game or educational software developer. This powerful program lets you create and edit 8X8 or 16X16 bit sprites; then, integrate them in your own BASIC language programs.

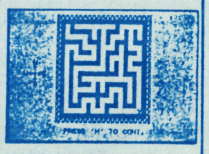


603D ADDRESS MASTER
ADDRESS MASTER lets you maintain and organize addresses of friends, acquaintances, business associates, and the like. You can create files, edit entries, sort addresses, search for an address, and even, print labels with this useful program.

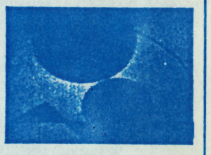
617T GRAPH MASTER
GRAPHMASTER turns your computer into a graphing machine. You can perform experiments in mathematics and geometry that will help you learn concepts taught in secondary school.



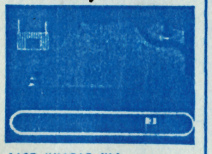
654T MORSE CODE TUTOR
If you have a desire to learn morse code, this is the program for you. With a few hours on MORSE CODE TUTOR, even the raw beginner will be ready to tap-out his own coded messages.



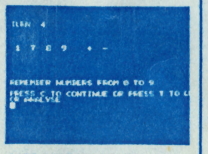
MAZE CUP CHAMPION SD244T
Challenge your sense of direction with this 3-dimensional maze.



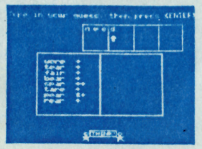
659T PAINTER
Here's a new creative outlet for your imagination. PAINTER let's you create and save your own colorful computer art. Seven powerful options help you paint lines, dots, boxes, circles, oval shapes and even, text.



663T JUNGLE JIM
Jim's bride has been kidnaped by man-eating natives! Help Jim as he makes his way through the treacherous jungle to save his sweetheart.

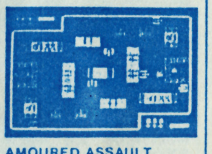


667T SPECTRABRAIN
The computer is thinking of a number. A number that could be in the tens, the hundreds, the thousands or even, the tens of thousands--can you guess it? There's no end to the fun and challenge of SPECTRABRAIN.



668T THINKI
Here's a real THINKing-man's game. The computer has selected a four letter word from it's 1600 word vocabulary--can you guess which one? Find out, play THINK.

669T PARA-JUMP
Gauge the wind speed and direction to land on the target and win your rightful place in the para-core.



AMOURED ASSAULT SD233C/T
Duel man against man or man against machine in this exciting and challenging World War II tank battle.

672T SUPERSAVER

A careless worker has ignited [illegible] and the whole oil rig is an inferno. Maneuver your helicopter to rescue the helpless workers before they are roasted by the flames.

JUNO LANDER SD306T

Use your skill as pilot to bring the cripple space craft to a smooth landing in this futuristic simulation.

312T PUZZLEBRICK

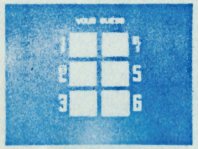
Rearrange numbers into proper order by shifting the blocks in the puzzle brick.

313T MASTERBRAIN

The computer is thinking of a sequence of numbers, can you guess them?

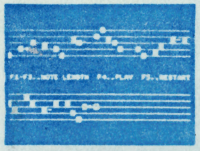
314T MOON LANDER

Fight gravity and a limited fuel supply to land your spacecraft safely.



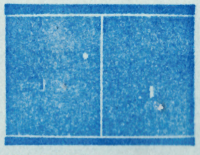
315T MEMORY TRAINER

How good is your recall? Find out with memory trainer.



316T MUSIC COMPOSER

An easy way to play and compose your own tunes.



317T TENNIS

Bring the challenge and excitement of a tennis tournament to your computer screen.

319T NOUGHTS & CROSSES

Can you out-smart the computer in a game of X and O's?

320T CHEST

Make clever use of clues to find the treasure in the least number of digs.

321T 21

Play the odds to beat the computer at a game of 21.

322T CALENDAR

Calendar, opens a new window to the future and past, enabling you to view a calendar of any month in the 19th, 20th, and 21st centuries.

SWING MAN SD221T

[illegible]

324T CALCULATOR

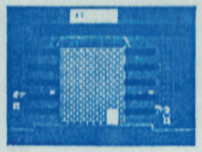
Turn your computer into a electronic calculator.

325T SQUASH

Basketball in this action packed game of computer squash.

326T HORSE RACE

Bet on the races then cheer your horse to victory as they race for the post, right before your eyes.

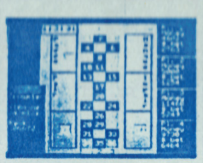


327T TROUBLE TROLLEY

Cooperate with a friend to put as many bricks as possible in storage above the factory.

SECTOR ALPHA SD220C

Defend Sector Alpha against robot invaders from the Throog Empire. Combine your careful aim with a periscope that can rotate a full 360 degrees and a powerful laser to defend every parameter.



330T ROULETTE

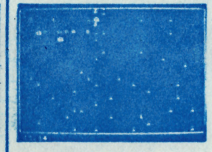
Place your bet. Spin the wheel. Will you win the pot or go bust? Find out with Roulette.

BIORHYTHM SD301T

[illegible]

FRANTIC FREDDY SC232C

Your cat is caught in a flaming inferno: you must help Frantic Freddy, the fearless fireman, put out the flames and save the day.

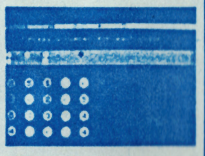


333T GRAVE DIGGER

Maneuver around tomb stones and avoid ghosts to escape from the grave yard before the clock strikes midnight.

FLIPPER SLIPPER SD291C

[illegible]

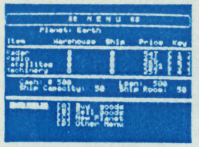


335T SPECTRAMIND

Find the hidden color sequence within 8 attempts.

336T PRINCE & DRAGON

Complete the number sequence to ascend the staircase, get the crown and save the princess.



337T SPECTRA TRADER

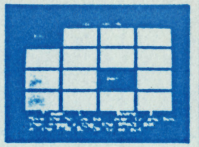
Rome the galaxy as an intersteller trader - buying and selling merchandise as you go.

338T PERIOUS JOURNEY

Become a wizard, thinker, barbarian, or warrior and start a thrilling and exciting journey.

NEW YORK BOMB BLITZ SD219T

Your engines are disabled as your ship enters the earths atmosphere. Falling rapid towards New York city, you must clear a landing site or be devastated by the towering sky scrappers.

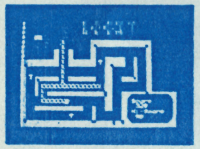


340T MATCH ME

Learn by playing matching games on English, Geography, Biology and Math.

SPECTRA SPRITE EDITOR SD266T

Sprite editor makes the creation and editing of sprites a powerful sprite in fast and easy process.



342T LOGIT

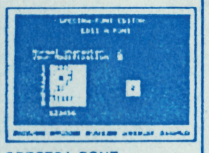
Make your way through the maze without getting caught by angry Sticks.

343T KIWI COUNTRY

Learn the names and locations of cities in New Zealand with the help of a high resolution map and an animated Kiwi bird.

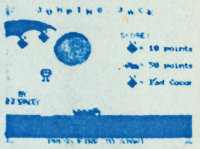
346T MATHS

A fun and easy way to drill your children in elementary arithmetic.



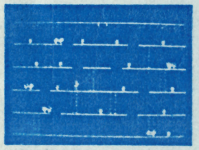
SPECTRA FONT EDITOR SD267T

[illegible]



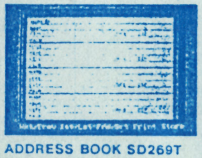
348T JUMPING JACK

Help Jack bound from one tree to the next to eat coconuts and collect points.



349T CRUNCH

Collect treasure by moving your man across the floors and up and down the ladders.

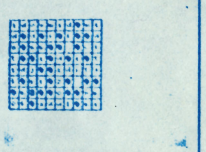


ADDRESS BOOK SD269T

[illegible]

351T PUZZLE MASTER

Reconstruct the picture from scrambled picture blocks.



353T SPECTRAWORD

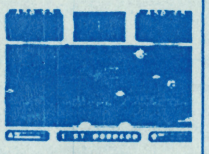
Discover the letter code to solve the crossword puzzle.

354T CAKE BANDIT

Answer math questions to help VIDI down the tree, so she can stop the cake bandit from stealing her cake.

219T NEW YORK BOMB BLITZ

Your engines are disabled as your ship enters the earths atmosphere. Falling rapid towards New York city, you must clear a landing site or be devastated by the towering sky scrappers.

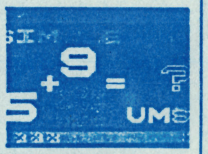


220C SECTOR ALPHA

Defend Sector Alpha against robot invaders from the Throog Empire. Combine your careful aim with a periscope that can rotate a full 360 degrees and a powerful laser to defend every parameter.

221T SWING MAN

Here's an exciting spelling game for your child. Help Swing Man swing from vine to vine through the dense jungle, by guessing the secret words.

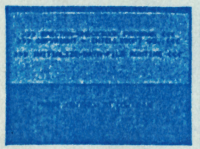


222T FIRST STEP

First Step is a fun and easy way to develop your child's letter and number recognition, and elementary arithmetic skills.

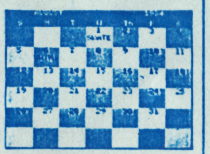
226T BUSY BEE

By manipulating Busy Bee with a joystick, even young children can create colorful computer art.



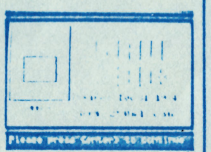
227T/D SPECTRA CHECKBOOK

This program offers a friendly and convenient tool for maintaining and updating your bank balance.



228T SPECTRA DIARY

Plan and detail your appointments, errands and important days to remember with this automated diary.

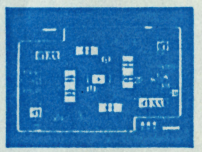


229T SPECTRA HOME ECONOMIST

Maintain and organize your mounting collection of discount and gift coupons with this coupon management system.

232C FRANTIC FREDDY

Your cat is caught in a flaming inferno: you must help Frantic Freddy, the fearless fireman, put out the flames and save the day.

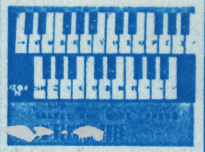


233T ARMOURED ASSAULT

Dual man against man or man against machine in this exciting and challenging World War II tank battle.

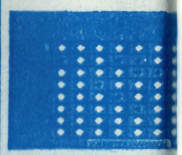
234T SPECTRON
Wather wave after wave of alien invaders to win the galactic wars and save your beloved motherland in this classic arcade game.

235T INTRODUCTION TO BASIC
Introduction to BASIC reveals the secrets of advanced BASIC programming enabling you to take full command of the Spectra's powerful graphics and sound capabilities.

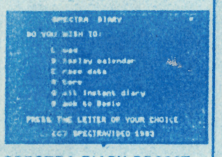


236C MUSIC MENTOR
Turn your computer into a musical instrument and create, edit and play your own synthesised music.

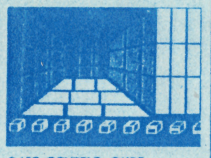
237C SUPER CROSS FORCE
Master the dial countersync laser-firing from two independent vantage points simultaneously-to out maneuver formidable fleets of "Marjul" ships and save the galaxy.



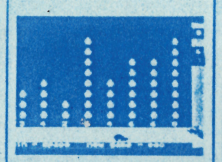
238T OTHELLO
The computer offers a clev and intelligent opponent i this classic game of strategy--Othello.



SPECTRA DIARY SD228T
Plan and detail your appointments, records and important days to forward with this automated diary.

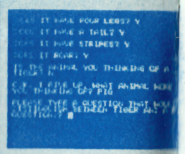


245T CRYPTIC CUBE
A computerized version of the popular rubic cube puzzle.



251T HARE & TORTOISE
Count objects to win the race between the tortoise and the hare.

252T FINDIT
Develop your vocabulary and practice your spelling with this automated derivative of the classic crossword puzzle.



253T SV. JUNGLE
Let the computer help teach your child some of the basi of biology by tailoring you own animal quiz.

SPECTRA CHECKBOOK SD227T
This program offers a friendly and efficient tool for maintaining and updating your bank balance.

255T NOMIS
Match wits with "Nomis" the master computer brian - test and develop your memory in this challenging number game.

256T STAR WORDS
Choose the letter that completes the word, displayed on your wordship, from hordes of "Mirrortron" imposters..

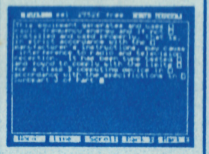
257T RESCUE
Avoid shells and radio aerials while fly over jagged peaks in your helicopter to save the three brave soldiers.

260T MINI GOLF
Turns your computer into a hole miniature golf course with enough challenge for a golf pro.

274T BONE OF CONTENTION
A game that will challenge and test your elementary math skills.

275T/D SPECTRA FILE CABINET
File Cabinet provides a quick and easy way to save and organise your valuable home information.

288T SPECTRA TYPE
Spectra Type is a fun way to improve your typing speed and accuracy.



294C JUSTWRITE JR.
JustWrite Junior is a userfriendly word processor specifically designed for light duty. Even the beginner can take full advantage of the easy editing and professional printing capabilities of this program.

296T SMASH
Bash the ball through layer after layer of bricks to sc points and break through th wall to win.

ACTION GAME
STRATEGY GAME
EDUCATIONAL PROGRAM
PROGRAMMING AIDS
PERSONAL INTEREST & BUSINESS APPLICATIONS

T = TAPE
C = CARTRIDGE
D = DISKETTE

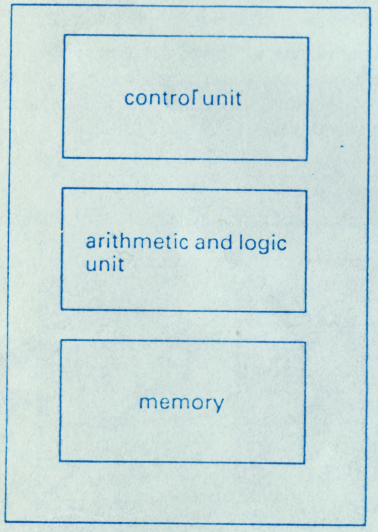
Powerful software for your Spectravideo or MSX computers available on mail order

COMPUTERS

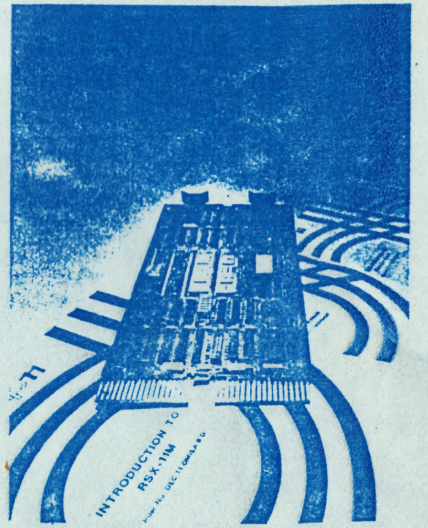
MORE ABOUT THE PARTS OF A COMPUTER

THE CENTRAL PROCESSING UNIT, OR CPU

The CPU is where processing of information takes place. It consists entirely of electronic components called **chips**, and has no moving parts. Some computers have their complete CPU in one chip. These chips are called **microprocessors**.
A CPU consists of three parts, as shown in the diagram below.



The parts of a central processing unit



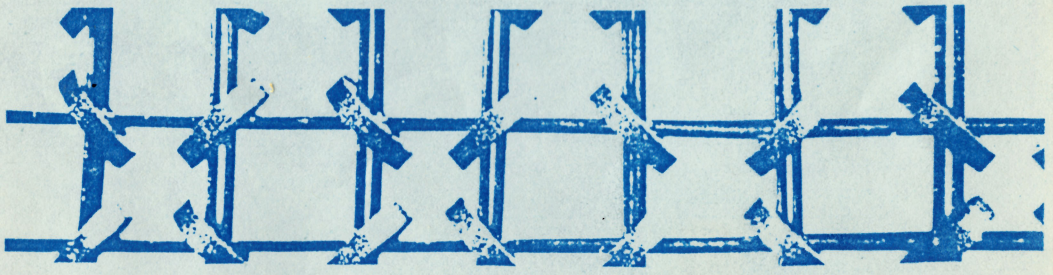
Part of the CPU of a microprocessor

The **memory**, or **main store** holds the programs and the data which a computer is using at the time. In older computers, the memory consisted of small metal rings, called **ferrite cores**. These can be magnetised in one direction or another to store a 0 or a 1.

In modern computers, the memory is made of electronic components like the rest of the CPU.

The memory of a computer consists of a number of **cells**. Each cell contains one item of data, or one program instruction. The contents of any memory cell can be examined, or replaced, at any time.

An old-fashioned computer memory made of ferrite cores



If we command our wealth, we shall be rich and free. If our wealth commands us, we are poor indeed.
—Edmund Burke

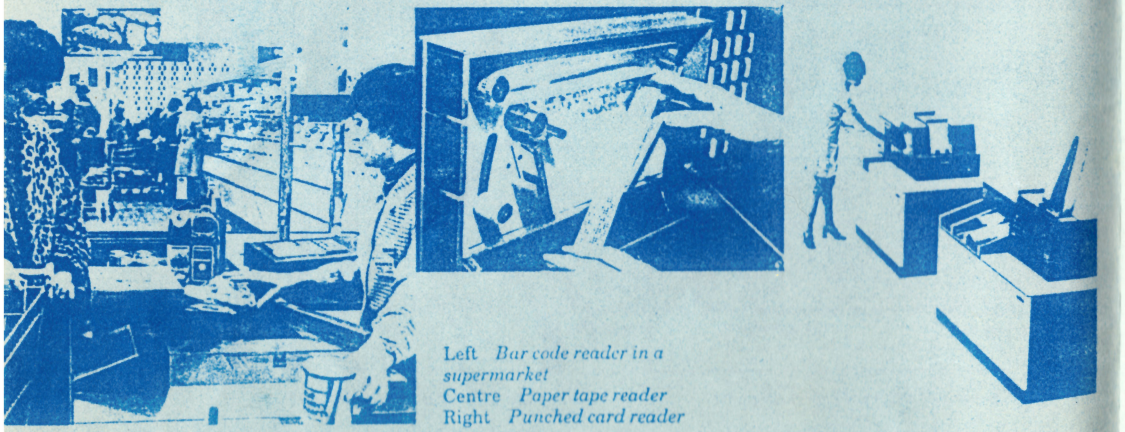
The **arithmetic and logic unit**, or **ALU**, is where calculations are done, decisions are made and other data processing operations are carried out.

The **control unit** controls the order in which program instructions are carried out. It is also responsible for the timing of all the operations done in the CPU.

INPUT DEVICES

Input devices read data or programs and feed them into a computer.

There are many different kinds of input devices. These include punched card readers, paper tape readers, bar code readers, magnetic ink character readers and special devices to read Kimball tags. Most computers have several input devices.



Left *Bar code reader in a supermarket*
Centre *Paper tape reader*
Right *Punched card reader*

A type of input method which is being developed at the moment is **voice recognition**. It may soon be possible to speak instructions or data to a computer.

Having read the input data, input devices send the data, now coded as 0's and 1's, to the CPU of the computer.

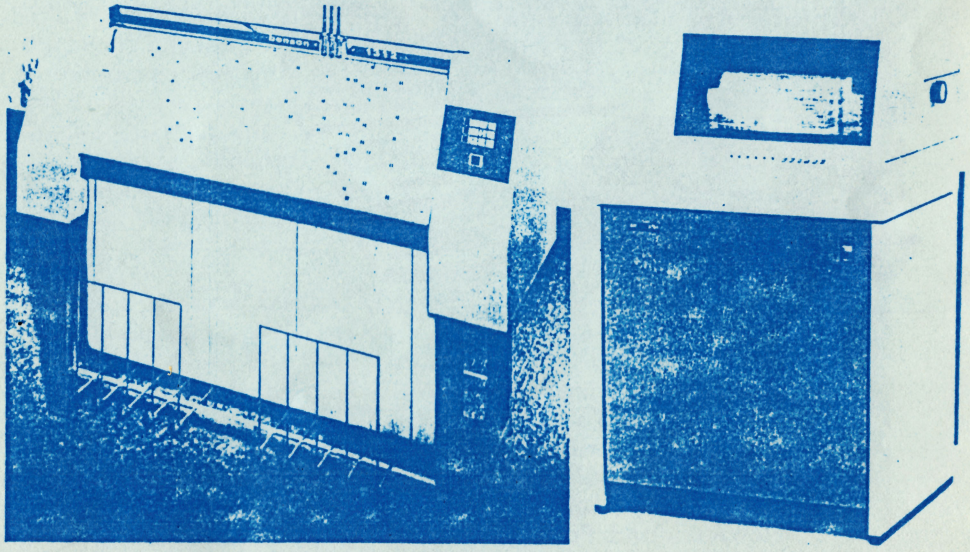
Right *Dr Fred Jelinek and Dr Alan Cole, two of the pioneers of computer voice recognition, experimenting with their equipment.*

OUTPUT DEVICES

Output devices print or display information which has been processed by a computer. The commonest output device is a **line printer**. A line printer can print data very quickly – one line at a time, in fact. Other output devices include visual display units, graph plotters and graphics display units. Graphical output is one of the fastest growing areas of computing.

Special cameras can be used to produce **computer output on microfilm (COM)**. A number of pages of output are reduced in size onto a single piece of film. A special microfilm reader is used to read the data. Computer output on microfilm saves large quantities of paper.

Top left *Graphics display unit*
Bottom left *Graph plotter*
Bottom right *Line printer*



The main store inside the CPU of a computer is usually too small to contain all the programs and data the computer might want to use. Extra space for programs and data is provided by **backing store**. The commonest backing store units are **magnetic tape units** and **magnetic disc drives**. Backing store can hold very large amounts of data.

One often hears of **data banks**, which are large amounts of information that can be accessed by computers. Data banks are kept on the backing store of computers.

OPERATOR'S CONSOLE

The operator's console is often attached to the CPU of a computer. It enables the computer to be started and stopped, programs to be loaded and run, and messages to be sent to the person operating the computer.

PUTTING THE UNITS TOGETHER

In medium and large sized computers, each unit is separate. The units are connected by a number of wires which run under the floor of the computer room. The computer can be expanded by adding new units, or modified by replacing units. Some very large computers have more than one CPU.

The smallest computers, called **microcomputers**, have a processing unit, attached to a keyboard and a visual display screen. How do these units do all the tasks mentioned earlier? The keyboard is the input device, and the operator's console. The screen shows input and output. The processing unit contains a microprocessor. If there is any backing store, it is a cassette tape recorder or floppy disc drive connected to the processing unit.

Microcomputer

Out of the *QE2*'s fall and rise has emerged a superliner with her own personality, and a ship's company with immense pride. In September last year the *QE2* sailed into an exceptionally fierce storm off Newfoundland. For 24 hours huge waves—some as high as 30 metres—flung the liner about.

After the ordeal, a woman passenger asked Captain Douglas Ridley: "Did you think of sending for help from the coast-guard?"

"Madam," replied the *QE2*'s commander, drawing himself up. "First, there was no need for help. Second, if there had been a coast-guard cutter in the area, it would be far more likely to have needed help than the *QE2*."

New colour Beltel software available for SVI and MSX

A colour software Beltel package is now available for the MSX and SVI 328 computers. It makes full use of the colour graphics screen used in these computers. The emphasis is placed on more facilities for the user.

The loading and saving of 32 pages, full editing and printing facilities and a local edit command means that you can create your own pages onto disk tape or computer memory without connection to the telephone system, save them and then later dial into Beltel and send these pages — one by one to Beltel.

Loading the programme

The tape based programme has a small "CLOAD" programme to get things started. Once this part is loaded, press run. This programme will call a second Basic programme that will immediately execute. This programme will then load the Assembler programme for Beltel as well as the information file. The information file contains all the towns and telephone numbers.

The Basic programme will then display the main menu. By selecting the desired town the number will automatically be dialled. When you choose function 5 or 6 you can alter the information file. This will allow you to change the town and telephone numbers. You can then save the new information back on tape by just following the instructions. Remember to cover the slot at the back of the cassette to allow you to record. Note that if you do not want to dial any number then you can just type a '-' in the place of the phone number and delete the rest.

After the number is dialled, listen on your audio output for the ringing and the Beltel answer tone. After a few seconds the screen will go blank, followed by the Beltel sign-on message.

If you have problems loading the tape, try again further along as the programme is recorded twice on the same side.

The software is available on cassette at R45.00 and R56.00 on disk

Spectravideo or MSX

Send your crossed cheques or postal orders to:
Greenstar National Computer Club
P.O. Box 3349
VANDERBIJLPARK 1900

BOOLEAN ALGEBRA FOR COMPUTER LOGIC

The 10^0 column contains units of ones, the 10^1 column is units of tens, 10^2 is units of hundreds, and 10^3 is units of thousands. Thus the number 3248 = 3000 + 200 + 40 + 8. The power is normally termed the *positional weight* of the number. Thus a 6 in the 10^3 column would represent 6000 while a 6 in the 10^0 column represents only 6. The least significant digit (lsd) is in the 10^0 column while the most significant digit (msd) is in the 10^3 column in this table.

Table 1-2 reveals the corresponding binary system for four bits. ("Bits" is a contraction of *bi*nary digi*ts*). Here, instead of ascending powers of ten, we have ascending powers of two. Only zeroes and ones are used. A 1 in the 2^0 column (least significant bit) is equal to 1. A 1 in the 2^1 column is equal to 2. A 1 in the 2^2 is equal to 4. A 1 in the 2^3 column is equal to 8.

Table 1-2. Binary Table for Four Bits

Binary Weight	2^3	2^2	2^1	2^0	
Decimal Equivalent	8	4	2	1	Decimal Count
	0	0	0	0	0
	0	0	0	1	1
	0	0	1	0	2
	0	0	1	1	3
	0	1	0	0	4
	0	1	0	1	5
	0	1	1	0	6
	0	1	1	1	7
	1	0	0	0	8
	1	0	0	1	9
	1	0	1	0	10
	1	0	1	1	11
	1	1	0	0	12
	1	1	0	1	13
	1	1	1	0	14
	1	1	1	1	15
	MSB			LSB	

To convert any 4-bit binary number to its equivalent decimal value, count from right to left and give each 1 its equivalent decimal value from the "weight" of its given position. For example, binary 0110 is simply 2 + 4 = 6. Binary 1001 is simply 1+8 = 9. Binary 1111 = 1+2+4+8 = 15.

Going Beyond Four Bits

Table 1-3 lists the positive powers of 2 for values up to 4096 decimal equivalent. Note that each additional power (weight) doubles the decimal equivalent value. Thus Tables 1-1 and 1-2 can be expanded to as many powers as required for a particular application. For example, binary 100101 = 32+4+1 = 37. Binary 1100101 = 64+32+4+1 = 101. Simply count the spaces to the left and give the digit the appropriate decimal equivalent, then add. In counting from right to left, just say 1-2-4-8-16-32-64-128-256, etc.

The Modulus

The odometer (mileage indicator) in your automobile normally has a decimal readout capacity of 99,999.9 miles. When you travel another 1/10 mile, the odometer "resets" or "clears" to 00,000.0, which actually represents 100,000 miles. Therefore,

Table 1-3. Positive Powers of 2

Power of 2	Decimal Equivalent
2^0	1
2^1	2
2^2	4
2^3	8
2^4	16
2^5	32
2^6	16
2^7	128
2^8	256
2^9	512
2^{10}	1024
2^{11}	2048
2^{12}	4096

for this device, the *modulus* is 100,000. The modulus of any counter is the fixed capacity of the counter *plus one* increment of the least significant digit (lsd).

Note from Table 1-2 that the 4-bit binary counts from 0 to 15. This is 16 "levels," including zero, and 16 is the modulus ($2^4 = 16$). Another way to look at this is to realize that the maximum decimal count is the modulus minus 1.

```
1 REM*********************************************
2 REM*     M S X     ONLY      MSX     ONLY    *
3 REM*                 HI - LO                 *
4 REM*       BY CLIFF WALTERS (JUNE 85)        *
5 REM*THIS PROGRAM HAS BEEN CHECKED &  *
6 REM*RUN ON A MITSUBISHI ML-F80 COMP.  *
7 REM* IT WORKS !!!        PLEASE.........*
8 REM* DO NOT CALL IN ON THE HOTLINE.  *
9 REM*********   IT DOES WORK !  ************
10 CLEAR500
20 ONSTOPGOSUB1520:STOPON
30 SCREEN1:WIDTH32:KEYOFF
40 DEFINTA-Z
50 LOCATE13,3:PRINT"HI-LOW",TAB(13)"******
**"
60 LOCATE8,20:PRINT"ONE  MOMENT  PLEASE"
70 N=RND(-TIME)
80 DIMC$(13,4)
90 DEFUSR=&H156
100 REM CHAR DEF
110 FORB=1TO8:FORA=0TO7
120 VPOKE(128+B)*8+A,VPEEK((49+B)*8+A)/2
130 VPOKE(146+B)*8+A,VPEEK((49+B)*8+A)/2
140 NEXTA,B
150 FORA=0TO7
160 VPOKE280+A,VPEEK(1248+A)
170 VPOKE1024+A,VPEEK(520+A)/2
180 VPOKE1168+A,VPEEK(520+A)/2
190 VPOKE1136+A,VPEEK(40+A)
200 VPOKE1144+A,VPEEK(48+A)
210 VPOKE1152+A,VPEEK(32+A)
220 VPOKE1160+A,VPEEK(24+A)
230 VPOKE1104+A,VPEEK(592+A)/2
240 VPOKE1248+A,VPEEK(592+A)/2
250 VPOKE1112+A,VPEEK(648+A)/2
260 VPOKE1256+A,VPEEK(648+A)/2
270 VPOKE1120+A,VPEEK(600+A)/2
280 VPOKE1264+A,VPEEK(600+A)/2
290 NEXTA
300 RESTORE350
310 C=0:READA:IFA=999THEN600
320 A$="":READA$:FORJ=1TOLEN(A$)STEP2
330 VPOKEA*8+C,VAL("&H"+MID$(A$,J,2))
340 C=C+1:NEXTJ:GOTO310
350 DATA96,007061420 50A1429
360 DATA97,7E99245AC324A5A5
370 DATA98,000E8642A0502894
380 DATA99,52E44241201C0201
390 DATA100,A5A5A524A57E1800
400 DATA101,4A2742820438 4080
410 DATA102,000000081C1C2A7F
420 DATA103,8142242424242424
430 DATA104,000000367F7F7F3E
440 DATA105,7f2A010E0E010008
450 DATA106,1800E71818E71818
460 DATA107,1C0880707080 0008
470 DATA108,1C3E7F7F3E08030E
```

THE GREATEST pleasure I know is to do a good action by stealth and have it found out by accident. – Charles Lamb

LOVE endures only when the lovers love many things together and not merely each other. – Walter Lippmann

```
480 DATA109,1818181818E78118
490 DATA110,1C3E7F3E1C08C070
500 DATA111,3841475C7000007 0
510 DATA112,7EC35ABD5A183C42
520 DATA113,1C86E23A0E00000E
530 DATA114,88A9AAB4A8906000
540 DATA115,C35A5A4224181800
550 DATA116,1195552D15090600
560 DATA137,0026692929297600
570 DATA155,0026692929297600
580 DATA160,FFC3BDA5A5BDC3FF
590 DATA999
600 D$=STRING$(5,8)+CHR$(10):E$=STRING$(
5,120):F$=""
610 FORI=1TO9:F$=F$+E$+D$:NEXTI:REM CARD
 BLANK
620 L$=STRING$(3,8)+CHR$(10):M$=STRING$(
3,160):N$=""
630 FORI=1TO7:N$=N$+M$+L$:NEXTI:REM CARD
 BACK
640 Q$="  ab"+L$+"cde"+L$+"fgh"+L$+"ijk"+
L$+"lmn"+L$+"opq"+L$+"rst":REM J,Q,K PIC
TURE
650 COLOR1,3,3:CL=8192
660 VPOKECL+12,177:VPOKECL+13,177:VPOKEC
L+14,177:VPOKECL+15,255:VPOKECL+16,31
670 VPOKECL+17,31:VPOKECL+18,143:VPOKECL
+19,143:VPOKECL+20,95:VPOKECL+31,0
680 REM DEAL
690 X(1)=1:X(2)=7:X(3)=13:X(4)=19:X(5)=2
5
700 CLS:GOSUB1450
710 M=100:GOSUB1390
720 FORT=1TO5
730 U=USR(0)
740 CARD(T)=INT(RND(1)*13)+1
750 SUIT(T)=INT(RND(1)*4)+1
760 IFC$(CARD(T),SUIT(T))="X"THEN740
770 C$(CARD(T),SUIT(T))="X"
780 IFT>1THEN810
790 LOCATE8,22:PRINT"PRESS  D  TO  DEAL"
800 GOTO820
810 LOCATE4,22:PRINT"HIGHER OR LOWER H O
R L?"
820 LOCATE28,22:K$=INPUT$(1)
830 K=ASC(K$):IFT<>1THEN850
840 IFK<>68THEN820ELSECLS:GOTO890
850 IFK=72THENLOCATE(T-1)*6+1,3:PRINT"HI
GH"
860 IFK=76THENLOCATE(T-1)*6+2,3:PRINT"LO
W"
870 IFK=72ORK=76THEN880ELSE820
880 LOCATE0,22:PRINTSTRING$(30,32)
890 GOSUB990
900 IFT=1THENLOCATE1,3:PRINT"DEALT"
910 IFT=1ORCARD(T)=1ORCARD(T-1)=1THEN960
920 IFK=72ANDCARD(T)>CARD(T-1)THEN970
```

POLITICS is like roller skating. You go partly where you want to go, and partly where the damned things take you. – Henry Fountain Ashurst

```
930 IFK=72ANDCARD(T)<CARD(T-1)THEN1380
940 IFK=76ANDCARD(T)>CARD(T-1)THEN1380
950 IFK=76ANDCARD(T)<CARD(T-1)THEN970
960 IFCARD(T)=CARD(T-1)THEN1380
970 NEXTT
980 GOTO1440
990 PLAY"T255L64V6o2B64o3B64o4B64o5B64"
1000 REM MAIN GRAPHIC ROUTINE
1010 G=SUIT(T)+141
1020 IFSUIT(T)<3THEN1050
1030 N=CARD(T)+145
1040 GOTO1060
1050 N=CARD(T)+127
1060 LOCATEX(T),6:PRINTF$
1070 LOCATEX(T),6:PRINTCHR$(N)
1080 LOCATEX(T)+4,14:PRINTCHR$(N)
1090 LOCATEX(T),7:PRINTCHR$(G)
1100 LOCATEX(T)+4,13:PRINTCHR$(G)
1110 ONCARD(T)GOSUB1190,1200,1220,1230,1
260,1270,1300,1320,1350,1360,1370,1370,1
370
1120 IFT>1THENRETURN
1130 XX=X(T)+6
1140 FORI=1TO4
1150 FORDL=1TO250:NEXTDL
1160 LOCATEXX,6:PRINTF$
1170 LOCATEXX+1,7:PRINTN$
1180 XX=XX+6:NEXTI:RETURN
1190 LOCATEX(T)+2,10:PRINTCHR$(G):RETURN
:REM ACE
1200 LOCATEX(T)+2,8:PRINTCHR$(G)
1210 LOCATEX(T)+2,12:PRINTCHR$(G):RETURN
:REM 2
1220 GOSUB1200:GOSUB1190:RETURN:REM 3
1230 FORP=1TO2
1240 LOCATEX(T)+2*P-1,8:PRINTCHR$(G)
1250 LOCATEX(T)+2*P-1,12:PRINTCHR$(G):NE
XTP:RETURN:REM 4
1260 GOSUB1190:GOSUB1230:RETURN:REM 5
1270 GOSUB1230
1280 LOCATEX(T)+1,10:PRINTCHR$(G)
1290 LOCATEX(T)+3,10:PRINTCHR$(G):RETURN
:REM 6
1300 GOSUB1270
1310 LOCATEX(T)+2,9:PRINTCHR$(G):RETURN:
REM 7
1320 FORP=1TO4
1330 LOCATEX(T)+1,5+2*P:PRINTCHR$(G)
1340 LOCATEX(T)+3,5+2*P:PRINTCHR$(G):NEX
TP:RETURN:REM 8
1350 GOSUB1320:GOSUB1190:RETURN:REM 9
1360 GOSUB1320:GOSUB1200:RETURN:REM 10
1370 LOCATEX(T)+1,7:PRINTQ$:RETURN:REM P
IC
1380 LOCATE12,20:PRINT"YOU LOSE":M=M-5*T
1390 LOCATE9,22:PRINT"YOU HAVE R";M
1400 FORDL=1TO750:NEXTDL
```

MAN STAGGERS through life yapped at by his reason, pulled and shoved by his appetites, whispered to by fears, beckoned by hopes.
—Eric Hoffer in *The Passionate State of Mind*

```
1410 FORF=1TO13:FORG=1TO4
1420 C$(F,G)="":NEXTG,F
1430 GOTO720
1440 LOCATE12,20:PRINT"YOU WIN":M=M+50:G
OTO1390
1450 PRINTSPC(10)"INSTRUCTIONS",SPC(10)"
**************":PRINT:PRINT
1460 PRINT"      YOU HAVE TO GUESS IF THE
":PRINT:PRINT"  NEXT CARD IS HIGHER OR L
OWER":PRINT:PRINT"      THAN THE PREVIOUS
 ONE.":PRINT:PRINT
1470 PRINT"       ACE IS HIGH OR LOW.":PR
INT:PRINT:PRINT"    YOU GET NOTHING FOR A
 PAIR":PRINT
1480 PRINT"  AND YOU START OFF WITH R 10
0.":PRINT:PRINT:PRINT" MAKE SURE THE CAP
S LOCK IS ON."
1490 LOCATE5,22:PRINT"PRESS SPACEBAR TO
PLAY"
1500 IFINKEY$<>" "THEN1500
1510 CLS:RETURN
1520 SCREENO:COLOR15,0,0
```

D U A L — F O R M A T

by CHRIS VAN DER WESTHUIZEN.....

```
1  ' Write machine code to data file
2  ' in hex, merge into basic program.
3  '
10 DEF FNHX$(X)=STRING$(2-LEN(HEX$(X)),"
0")+HEX$(X)
20 INPUT "START LINE NO.";LN
30 INPUT "INCREMENT";IC
40 INPUT"FILE TO WRITE TO";FL$
50 INPUT"START ADDRESS  (HEX)";SA$
60 INPUT"END ADDRESS    (HEX)";EA$
70 SA=VAL("&H"+SA$)
80 EA=VAL("&H"+EA$)
90 OPEN FL$  FOR OUTPUT AS#1
100 FOR AD=SA TO EA
110 IF ADMOD8=0 THEN PRINT #1,LN;"DATA "
;:LN=LN+IC
120 PRINT #1,FNHX$(PEEK(AD));
130 IF ADMOD8=-1 THEN PRINT #1,"" ELSE P
RINT #1,",";
140 NEXT AD
```

A Computer for Today—and All Your Tomorrows...

MITSUBISHI
MSX COMPUTER SYSTEMS

Spectravideo 318 / 328

WE THANK 'SMUG' OF MONTREAL CANADA FOR THIS PROGRAM. THE PROGRAM HAS NOT BEEN CHECKED. IF YOU FIND ANY BUGS, PLEASE LET US KNOW.

```
10 '
20 '
30 '        Planet-x:1984
40 '        By Caspar Ryan
50 '        Uses 5K bytes Approx.
60 '
70 '        15/12/83
80 '
90 CLEAR 100
100 PLAY"v5o614e-05r64l64gr64gl12b-
14ar64af","o414gr64l64o3e-r64e-112g
e-14e-r64e-","v5o314b-r64l64b-r64b-
r64b-o4e-o314b-c-b-"
110 COLOR 15,1,1:SCREEN 2:FOR Y=1 T
O 12
120 FOR T=1 TO 8:R(T)=2+INT(RND(6)*
13):NEXT T:LOCATE 30,50:COLOR R(1),
1:PRINT"P";:COLOR R(2),1:PRINT"L";:
COLOR R(3),1:PRINT"A";:COLOR R(4),1
:PRINT"N";:COLOR R(5),1:PRINT"E";:C
OLOR R(6),1:PRINT"T";:COLOR R(7),1:
PRINT"-";:COLOR R(8),1:PRINT"X":NEX
T Y
130 SCREEN 1,2
140 PLAY"v6o514e-r64l64gr64gl12b-14
ar64af","o414gr64l64o3e-r64e-112ge-
14e-r64e-","v6o314b-r64l64b-r64b-r6
4b-o4e-o314b-c-b-"
150 STOP ON:ON STOP GOSUB 870:F=0:R
ESTORE 880:FOR I=1 TO 8:READ C:Z$=Z
$+CHR$(C):NEXT:SPRITE$(1)=Z$
160 FOR H=1 TO 8:READ PB:S$=S$+CHR$
(PB):NEXT H:SPRITE$(A)=S$
170 DEFINT A-Z:ON SPRITE GOSUB 460
180 UNSTOPGOSUB870:STOPON:CLICKOFF
190 ON INTERVAL=100GOSUB 910
200 CLS:FOR T=1 TO 80:Q=INT(RND(1)*
240):P=INT(RND(2)*168):PSET(Q,P),15
:NEXT
210 CIRCLE(60,30),25,11,,,1.25:CIRC
LE(152,22),8,7,,,1.2:CIRCLE(240,10)
,45,3,,,1.2
220 LINE(1,175)-(20,155),8:LINE(20,
155)-(40,175),8:LINE(40,175)-(50,17
5),8:LINE(50,175)-(60,165),8:LINE(6
0,165)-(70,175),8:LINE(70,175)-(120
,175),8:LINE(120,175)-(160,135),8
230 LINE(120,175)-(160,190),8:LINE(
159,135)-(159,189),8:LINE(160,135)-
(160,190),9:LINE(120,175)-(160,190)
,8:LINE(160,135)-(200,175),9
240 LINE(159,190)-(200,175),9:LINE(
200,175)-(210,165),8:LINE(210,165)-
(220,175),8:LINE(220,175)-(225,170)
,8:LINE(225,170)-(230,175),8
250 LINE(225,170)-(230,175),8:LINE(
230,175)-(255,175),8
260 PAINT(60,30),11:PAINT(152,22),7
:PAINT(240,10),3:PAINT(165,170),9:P
AINT(150,150),8
270 CIRCLE(245,10),4,1:CIRCLE(250,2
5),3,1:CIRCLE(220,5),3,1:CIRCLE(225
,28),4,1:CIRCLE(250,45),4,1:CIRCLE(
240,35),3,1:CIRCLE(235,45),3,1:CIRC
LE(220,40),2,1:CIRCLE(224,14),3,1:C
IRCLE(235,20),4,1:CIRCLE(235,5),2,1
:CIRCLE(210,12),4,1:CIRCLE(213,24),
2,1
280 CIRCLE(64,40),5,1:PSET(64,40),1
:LINE(45,175)-(45,183),15:LINE(40,1
75)-(50,175),15:LINE(95,175)-(95,18
3),15:LINE(90,175)-(100,175),15:LIN
E(242.5,175)-(242.5,183),15:LINE(23
7.5,175)-(247.5,175),15
290 PLAY"v6T255L603CFA04CR6L603A04L
2CR":PLAY"v6T255L604CFA05CR6L604A05
L2C":LOCATE 90,1:PRINT"PLANET-X":SP
RITE ON:INTERVAL ON
300 Q=128:P=10:PUT SPRITE 1,(Q,P),1
5,A:FOR T=1 TO 2500:NEXT T
310 R=R+4:PUT SPRITE 13,(R,168),15,
1:X=X-3:PUT SPRITE 0,(X+128,79),14,
1:X=X-3:PUT SPRITE 2,(X+60,70),14,1
:X=X-3:PUT SPRITE 3,(X,75),14,1:M=M
+4:PUT SPRITE 4,(M,95),15,1:M=M+4:P
UT SPRITE 5,(M+128,90),15,1:M=M+4:P
UT SPRITE 15,(M+60,100),15,1
320 C=C-5:PUT SPRITE 6,(C,116),14,1
:C=C-5:PUT SPRITE 7,(C+90,112),15,1
:D=D+4:PUT SPRITE 8,(D,140),14,1:D=
D+4:PUT SPRITE 9,(D+68,130),15,1:D=
D+4:PUT SPRITE 10,(D+155,138),14,1
330 K=K+4:PUT SPRITE 14,(K+100,168)
,15,1:W=W-6:PUT SPRITE 16,(W+251,15
0),15,1:P=P+1:PUT SPRITE 1,(Q,P),15
,A
340 IF STRIG(0)OR STRIG(1)THEN BEEP
:F=F+30:LET Q=5+INT(245*RND(-TIME))
:LET P=INT(165*RND(-TIME)):PUT SPRI
TE 1,(Q,P),15,A:IF F)400 THEN 440
350 JS=STICK(0)+STICK(1)
360 IF JS=1 THEN P=P-8:PUT SPRITE 1
,(Q,P),15,A:F=F+1:IF F)400 THEN 440
370 IF JS=3 THEN Q=Q+1:PUT SPRITE 1
,(Q,P),15,A:F=F+3:IF F)400 THEN 440
380 IF JS=7 THEN Q=Q-1:PUT SPRITE 1
,(Q,P),15,A:F=F+3:IF F)400 THEN 440
390 IF JS=5 THEN P=P+2:PUT SPRITE 1
,(Q,P),15,A:F=F+2:IF F)400 THEN 440
400 IF P=166 OR P=167 THEN 540
410 IF P)175 THEN 450
420 SP=SP+3:PUT SPRITE 25,(SP,10),1
2,A
430 GOTO 310
440 SPRITE OFF:LOCATE 65,95:PRINT"F
UEL SUPPLY DEPLETED!":FOR T=P TO 16
5:P=P+1:PUT SPRITE 1,(Q,P),15,A:NEX
T:GOTO 460
450 LOCATE 25,95:PRINT"YOU HIT THE
GROUND AT HIGH SPEED!!"
460 PUT SPRITE 1,(Q,P),8,A:SOUND 7,
8
470 INTERVAL OFF:FOR Y=1 TO 10:CIRC
LE(Q,P),Y,9:SOUND 6,15:SOUND 7,7:SO
UND 8,16:SOUND 9,16:SOUND 10,16:SOU
ND 10,16:SOUND 12,16:SOUND 13,0:NEX
T Y
480 LOCATE 65,75:PRINT"PLANET-X IS
DESTROYED!"
490 SOUND 7,16:FOR EX=1 TO 16:SOUND
 7,16-EX:COLOR,,15:FOR Z=1 TO 20:NE
XT:COLOR,,1:FOR Z=1 TO 5:NEXT Z,EX
500 SOUND 7,1:SOUND 12,255:SOUND 11
,255:SOUND 13,14:FOR EX=1 TO 16:SOU
ND 8,19:FOR X=1 TO 20:COLOR,,15:FOR
 Z=1 TO 20:NEXT:COLOR,,1:FOR Z=1 TO
 5:NEXT Z,EX
510 SOUND 7,8:PLAY"v7T11003L4DR64DL
64R64DL4R64DR64FL60R64EL6R64ER64L6D
R64DR64C#R64D"
520 LINE(160,135)-(240,15),5:LINE(1
60,135)-(250,20),5:LINE(250,20)-(24
0,15),5
530 FOR U=1 TO 60:CIRCLE(240,10),U,
13:SOUND7,10:SOUND9,16-UMOD16:NEXT:
FOR T=1 TO 2000:NEXT:SOUND9,0:GOTO
810
540 IF Q=40 OR Q=41 OR Q=42 OR Q=90
 OR Q=91 OR Q=92 OR Q=238 OR Q=239
OR Q=240 THEN 560
550 GOTO 460
560 INTERVAL OFF:RESTORE 900:FOR I=
1TO32:READ TA:TA$=TA$+CHR$(TA):NEXT
 I:SPRITE$(19)=TA$
570 'play""
580 SPRITE OFF:FOR ZZ=255 TO Q+10 S
TEP -1:PUT SPRITE 30,(ZZ,175),14,19
:FOR I=1 TO 40:NEXT I:NEXT ZZ:SPRIT
E ON
590 PLAY"AAADEEDFDFDF":INTERVAL ON
600 FOR T=1 TO 2000:NEXT T
610 P=P-1:PUT SPRITE 1,(Q,P),15,A
620 IF P=39 OR P=40 OR P=41 THEN 74
0
630 IF STRIG(0) OR STRIG(1) THEN BE
EP:LET Q=5+INT(245*RND(-TIME)):LET
P=1+INT(160*RND(-TIME)):PUT SPRITE
1,(Q,P),15,A:F=F+30:IF F)400 THEN 4
40
640 AD=STICK(0)+STICK(1)
650 IF AD=5 THEN P=P+8:PUT SPRITE 1
,(Q,P),15,A:F=F+2:IF F)400 THEN 440
660 IF AD=7 THEN Q=Q-1:PUT SPRITE 1
,(Q,P),15,A:F=F+2:IF F)400 THEN 440
670 IF AD=3 THEN Q=Q+1:PUT SPRITE 1
,(Q,P),15,A:IF P=40 THEN GOTO 740:F
=F+2:IF F)400 THEN 440
680 IF AD=1 THEN P=P-3:PUT SPRITE 1
,(Q,P),15,A:F=F+5:IF F)400 THEN 440
690 X=X+3:PUT SPRITE 0,(X+128,70),1
4:X=X+3:PUT SPRITE 2,(X+60,70),15:X
=X+3:PUT SPRITE 3,(X,70),15:M=M-4:P
UT SPRITE 4,(M,90),14:M=M-4:PUT SPR
ITE 5,(M+60,90),15:M=M-4:PUT SPRITE
 6,(M+128,90),15
700 C=C+3:PUT SPRITE 7,(C,110),14:C
=C+3:PUT SPRITE 8,(C+90,110),14:C=C
+3:PUT SPRITE 9,(C+170,110),14:D=D-
3:PUT SPRITE 10,(D,130),15:D=D-3:PU
T SPRITE 11,(D+110,130),15:D=D-4:PU
T SPRITE 12,(D+120,130),15
710 K=K+4:PUT SPRITE 16,(K+100,160)
,14,1:W=W+4:PUT SPRITE 15,(W+190,16
0),15:KK=KK-1:PUT SPRITE 13,(KK,60)
,15,1:PUT SPRITE 14,(KK+100,60),15,
1
720 SP=SP-3:PUT SPRITE 25,(SP,10),1
2,A
730 GOTO 610
740 IF Q=59 OR Q=60 OR Q=61 THEN 76
0 ELSE SPRITE OFF:FOR Q=1 TO 400:PU
T SPRITE 1,STEP(4,-1),15,A:NEXT:LOC
ATE 50,60:PRINT"YOU WERE SUCKED INT
O SPACE":GOTO 470
750 IF Q=60 THEN 760
760 INTERVAL OFF:LOCATE 80,80:PRINT
"YOUR WORLD IS SAVED!"
770 LOCATE 80,90:PRINT"CONGRATULATI
ONS"
780 LINE(64,40)-(160,135),6:LINE(14
0,150)-(150,160),1:LINE(150,160)-(1
70,150),1:LINE(170,150)-(160,130),1
:LINE(160,130)-(140,150),1:PAINT(15
5,155),1:A=A+1000:SC=A-F
790 LOCATE 60,65:PRINT"YOUR SCORE I
S:"SC:FOR T=1 TO 3000:NEXT T:RUN 15
0
800 FOR J=1 TO 1500:NEXT:GOTO 810
810 CLS
820 SCREEN 1:LOCATE 90,94:PRINT"TRY
 AGAIN(Y/N)"
830 AC$=INPUT$(1)
840 IF AC$="Y" OR AC$="y"THEN RUN 1
50
850 IF AC$="N" OR AC$="n"THEN END
860 GOTO 830
870 CLEAR:SOUND 8,0:RUN 150
880 DATA60,255,127,254,252,127,127,
44,16,56,16,84,84,124,84,238,60
890 DATA 16,56,16,84,84,124,84,238,
60
900 DATA 0,0,0,0,0,0,130,254,135,1,0,
1,255,145,68,36,31,0,0,0,4,200,200,
88,248,224,196,248,255,17,66,68,248
910 PLAY"v10t255o2l64fgfg"
920 RETURN
```


Celestial BASIC

coming soon !

Spectravideo 318 / 328

```
100  '  SV.328 HI-RES SCREEN-DUMP
110  '  TO EPSON RX-80 PRINTER
120  '  BY CHRIS V D WESTHUIZEN
130  '
140  '     ******* TYPE WIDTH 40 ******
150  '  THIS PROGRAMME PRINTS THE PICTURE
160  '  ON THE HI-RES SCREEN  SIDEWAYS ON
170  '  THE PRINTER.  IF YOU HAVE  ANOTHER
180  '  PRINTER, MAKE SURE IT IS COMPATI-
190  '  BLE.  RUN THE  PROGRAM,  THEN  RUN
200  '  YOUR  PROGRAMME  THAT  DRAWS  THE
210  '  PICTURE.
220  '  USE  DEF USR=&HD480 : Z=USR(0) TO
230  '  START THE ACTION.
240  '  NOTE THAT THE PROGRAMME PRINTS  A
250  '  3-BY-4 MATRIX FOR EACH PIXEL.  YOU
260  '  CAN RE-DEFINE THE PATTERN PRINTED
270  '  FOR EACH COLOR. LINES  780 TO 850
280  '  HOLDS 4 BYTES FOR EACH COLOR.  THE
290  '  EASIEST WAY IS TO PLACE THE BYTES
300  '  IN BINARY FORM IN DATA STATEMENTS
310  '  AND POKE  IT  INTO  THE  RELEVANT
320  '  LOCATIONS- FROM &HD560 TO &HD59F.
330  '
340  CLEAR 200,&HD480
350  FOR AD=&HD480 TO &HD59F
360  READ H$:V=VAL("&H"+H$):POKE AD,V
370  NEXT AD
500  DATA F3,3E,1B,CD,4E,D5,3E,40
510  DATA CD,4E,D5,3E,1B,CD,4E,D5
520  DATA 3E,30,CD,4E,D5,3E,0A,CD
530  DATA 4E,D5,3E,1B,CD,4E,D5,3E
540  DATA 41,CD,4E,D5,3E,03,CD,4E
550  DATA D5,01,FF,FF,11,C0,00,ED
560  DATA 43,58,D5,ED,53,5A,D5,03
570  DATA 1B,CD,E9,48,ED,4B,58,D5
580  DATA 03,11,C0,00,ED,53,5A,D5
590  DATA ED,43,58,D5,3E,1B,CD,4E
600  DATA D5,3E,4C,CD,4E,D5,3E,00
610  DATA CD,4E,D5,3E,03,CD,4E,D5
620  DATA ED,5B,5A,D5,1B,ED,53,5A
630  DATA D5,C5,D5,CD,E9,48,ED,4B
640  DATA 58,D5,ED,5B,5A,D5,CD,51
650  DATA 49,A7,20,03,3A,0C,FA,CD
660  DATA 2A,D5,ED,4B,58,D5,ED,5B
670  DATA 5A,D5,CD,CF,49,D1,C1,3E
680  DATA 00,BB,20,CC,CD,59,4A,3E
690  DATA 0D,C5,CD,4E,D5,3E,0A,CD
700  DATA 4E,D5,C1,3E,FF,B9,20,94
710  DATA FB,C9,C5,D5,E5,87,87,21
720  DATA 60,D5,11,00,00,5F,19,7E
730  DATA CD,4E,D5,23,7E,CD,4E,D5
740  DATA 23,7E,CD,4E,D5,23,7E,CD
750  DATA 4E,D5,E1,D1,C1,C9,C5,D5
760  DATA 37,3F,CD,15,39,D1,C1,C9
770  DATA 00,00,00,00,00,00,00,00
780  DATA 00,00,00,00,07,07,07,07
790  DATA 02,07,07,02,00,02,02,00
800  DATA 02,07,07,02,02,05,05,02
810  DATA 02,07,07,02,00,02,02,00
820  DATA 02,05,05,02,00,02,02,00
830  DATA 04,07,07,01,04,02,02,01
840  DATA 02,07,07,02,05,00,00,05
850  DATA 07,00,00,07,00,00,00,00
```

NEW

Toys for tomorrow from Quickshot™

Quickshot, who brought you the famous joysticks for your Spectravideo computers, have now developed a range of toys, **the Parabolic Ear, the Voice Modulator** and **the Robotarm.**

The Parabolic Ear which has already arrived in South Africa can pick up sound emitted from a target as far as 20 feet away by aiming the parabolic dish in the direction desired. The package includes headphones, a parabolic dish, and a microphone gun with a built-in amplifier.

NEW

Greenstar National Computer Club
P.O. Box 3349
VANDERBIJLPARK 1900

LOG TO ANY BASE

The logarithm of any number is the power to which the base of the logarithm must be raised to produce the number. This is a classic definition of logarithms. Generally, only two bases are used to calculate logarithms. The base 10 and the base e (where e = 2.71828). The latter is often referred to as the natural logarithm.

Sometimes to make calculations easier, it is desirable to calculate logarithms to bases other than these. In addition, most BASIC interpreters used in home computing systems do not even provide the elementary log-to-base-10 capability. They only permit calculation of the natural log (often written "ln").

This program makes it possible for your computer to calculate the log of any number to any base. For example, the log of 64 to the base 4 equals 3 and ln e equals 1. This can be helpful in computer applications for calculating what power of 2 a particular number is. For example, if the number 65536 is entered when this program is run and the base is 2, we discover that 65536 is equal to 2 to the 16th power.

The calculation is based on the definition of a logarithm stated above. From this definition, the following formula can be derived:

$$base \uparrow log = number$$

where the number is produced by raising the base to the log power. If the logarithm of both sides of this equation is taken, the result is

$$\log(base \uparrow log) = \log(number) \quad \text{or} \quad \log(B \uparrow L) = \log n$$

The left side of this last equation, however, is equal to $L * \log(B)$ so that the final result is

$$L * \log B = \log n$$

If both sides of the equation are divided by log B, the formula used by the program to evaluate the log of any number to any base results. This formula is

$$L = (\log n)/(\log B)$$

```
1  PRINT : PRINT : PRINT : PRINT "WRITTEN BY JULES H. GILDER"
2  PRINT : PRINT
3  FOR X = 1 TO 4000: NEXT X
10  PRINT : PRINT : PRINT : PRINT
20  PRINT "            LOG TO ANY BASE"
30  PRINT : PRINT : PRINT : PRINT
40  PRINT "THIS PROGRAM WILL CALCULATE THE LOGA-"
50  PRINT "RITHM OF ANY NUMBER TO ANY BASE."
60  PRINT : PRINT
70  INPUT "ENTER THE NUMBER  ";N
80  INPUT "ENTER THE BASE  ";B
90  PRINT : PRINT : PRINT
100 LN =  LOG (N) /  LOG (B)
110  PRINT "THE LOG OF ";N;" TO THE BASE ";B
120  PRINT
130  PRINT "IS EQUAL TO ";LN
140  END
```

```
ENTER THE NUMBER  65536
ENTER THE BASE  2

THE LOG OF 65536 TO THE BASE 2
IS EQUAL TO 16
```

DIE WET VAN OHM

Die Towerdriehoek

Oorweeg nou enige kring waarin die waardes van enige twee van die drie faktore—spanning, stroom en weerstand—bekend is, en waarin u nou die derde wil bepaal.

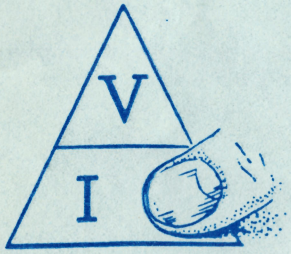
Die reël vir die werking van die driehoek is soos volg:

Plaas u duim oor die letter in die driehoek waarvan die waarde bepaal moet word—en die formule vir die berekening van die waarde word deur die ander twee letters aangegee.

Hier is hoe hierdie nuttige hulpmiddel in die praktyk werk:

1. U weet wat die waardes van stroom en weerstand in 'n kring is, maar u het nie 'n voltmeter tot u beskikking om die spanning te meet nie.

Teken maar die towerdriehoek; plaas u duim op die waarde wat u wil bereken (d.w.s. V) en wat oorbly, is die formule wat u nodig het—**I maal R.**

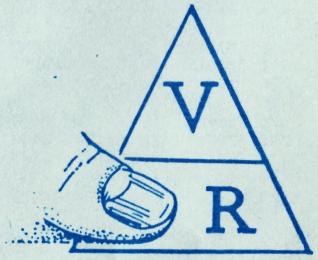


2. U weet wat die waardes van stroom en spanning is, maar u het geen ohmmeter tot u beskikking om die weerstand te meet nie. Plaas u duim oor die letter R; en wat oorbly is die formule $\frac{V}{I}$

Vervang die simbole met die bekende waardes vir V en I en die antwoord sal R wees.

3. Die spanning en weerstand van 'n kring is aan u bekend, maar die ammeter wat nodig is om die stroom te meet, is stukkend of het verlore geraak.

Plaas u duim oor die simbool vir stroom (I) en u kry die formule wat u nodig het: $\frac{V}{R}$



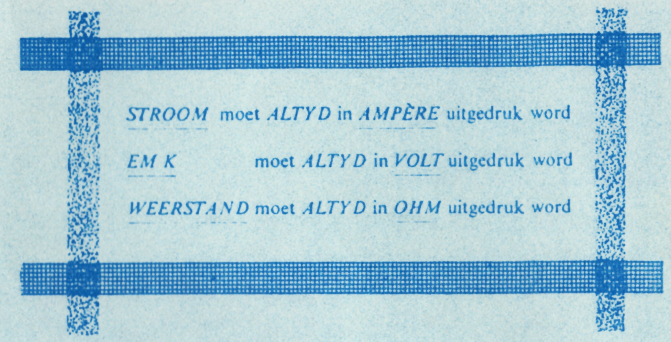
So ver so goed. As u die eenvoudige reël onthou: *bedek die waarde wat u nie weet nie.*

Maar as u 'n bietjie daaroor nadink, sal u besef dat die formule met betrekking tot die Wet van Ohm nie behoorlik kan werk tensy alle waardes in hul korrekte *maateenhede* uitgedruk word nie.

Reëls met Betrekking tot die Wet van Ohm

Die Eerste en Belangrikste Reël

Die Wet van Ohm sal alleenlik reg werk en korrekte antwoorde op 'n probleem verstrek, wat u met behulp daarvan wil oplos, as u die volgende onthou met betrekking tot die vergelyking van die Wet van Ohm.



Neem 'n kring waarin die weerstandsmeting 10 ohm is, en die stroommeting 300 milliampère is. Klaarblyklik, as u die Wet van Ohm-vergelyking blindelings volg, en eenvoudig neerskryf dat V = I × R = 10 × 300 = 3000, sal die spanning met 'n faktor van 1 000 verkeerd wees.

Hier moet u eintlik die herleidingstabelle gebruik, waarvan u in *Deel 1* geleer het (onthou u nog al daardie herleidingsoefeninge wat u moes deurwerk?), en u moet al die faktore in die eenvoudige vergelyking hierbo in ampère, volt en ohm herskryf.

U kry dus die gewysigde vergelyking;

wat vir u die korrekte antwoord gee. $V = I \times R = 10 \times ,3 = 3$

Die Tweede Reël

Daar is een reël waarby u altyd sal baat, as u dit vanuit die staanspoor toepas, elke keer wanneer u 'n probleem aangaande die Wet van Ohm probeer oplos, waarby groothede en waardes in 'n elektriese kring betrokke is. Dit is soos volg: **Teken altyd 'n ruwe diagram van die kring onder oorweging, voordat u met die berekeninge begin, wat die waardes in die kring, wat reeds aan u bekend is, as grondslag het.**

U sal uitvind dat hierdie reël mettertyd gebiedend noodsaaklik word, want niemand behalwe miskien 'n genie kan eintlik 'n ingewikkelde elektriese kring ontleed, sonder om eers 'n sketsie van die kringdiagram te maak nie, met die reeds bekende waardes daarop aangebring en daarna die ander toevoeg, soos hulle deur middel van berekening vanaf die oorspronklike gegewens verkry word.

Hoe gouer u dus gewoond daaraan raak om altyd 'n skets te teken van 'n Ohm se Wet-kring, voordat u dit probeer oplos, hoe beter.

Egg Beater

WHILE clipping the hedge, I listened to two brothers at play. "I'll give you two rands if you let me break three eggs on your head," said the elder one.

"Word of honour?" asked the younger one suspiciously.

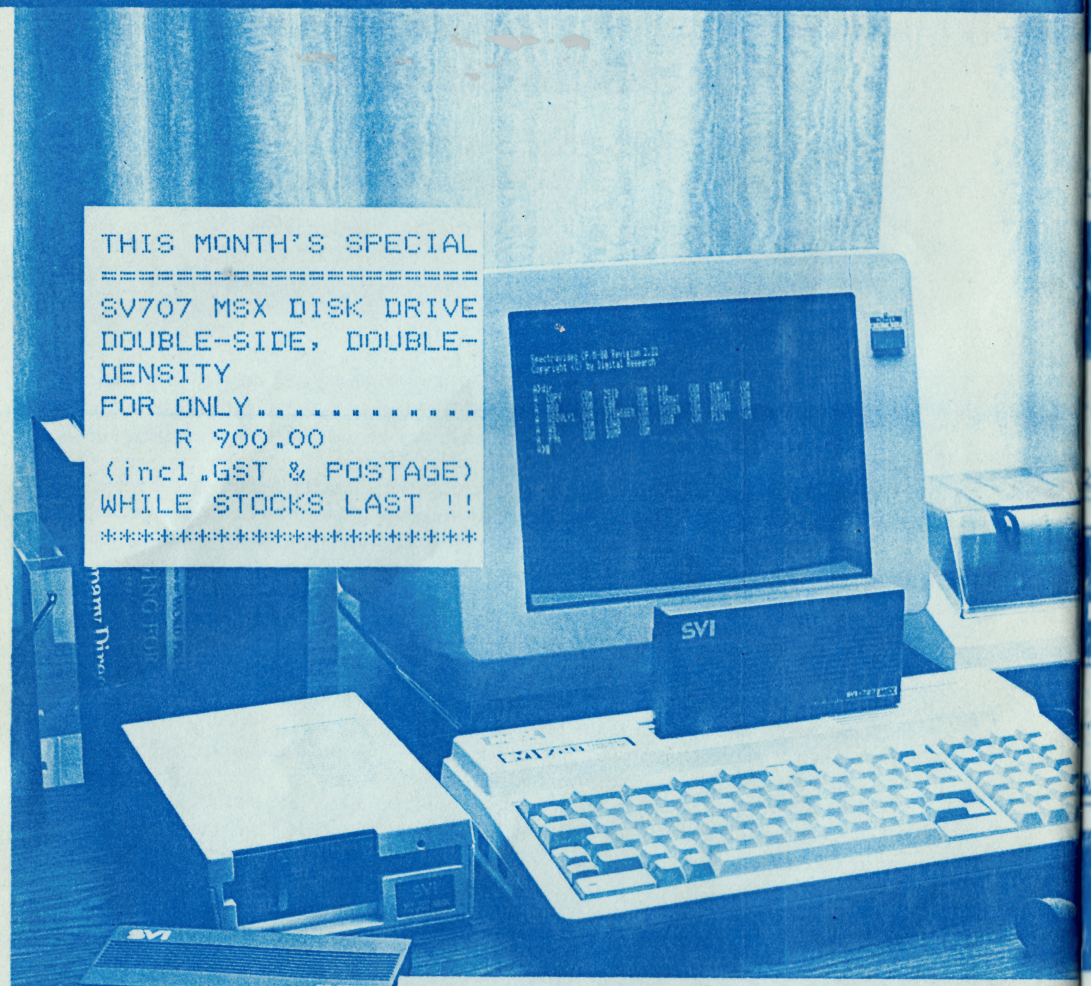
"Word of honour," answered the other.

Shaking with laughter, the elder boy broke the first two eggs over his brother's head. After standing as stiff as a ramrod for fear that the slimy mess would get on his clothes, the younger brother cried, "Well, when is the third egg coming?"

"I'm not stupid," replied his brother. "That would cost me two rands."

—Els Hoogmans

The SVI·728 will grow on you.



The Home Management System

SVI-707 MSX DISK DRIVE

An intelligent, double-sided, double density drive with a large formatted storage capacity of 320K bytes. The built-in controller can plug into any MSX computer. The SVI-707 is designed to run MSX-DOS and other MSX compatible operating systems.

SVI-727 MSX 80 COLUMN CARTRIDGE

Plugs into the cartridge slot

SVI·707

Nog toevoereenhede

Dit maak aan die mikrorekenaar geen verskil hoe dit inligting kry nie. Dit is egter nuttig as dit 'n toevoerpoort ('n spesiale sok) het sodat die toevoereenheid in die regte deel van die mikrorekenaar ingeprop kan word.

Dieselfde kodeerstelsel

Dit is ook noodsaaklik dat die mikrorekenaar en die toevoereenheid dieselfde kodeerstelsel gebruik. Die toevoereenheid kan byvoorbeeld nie 1001110 sein om 'n punt ("full stop") aan te dui, as die mikrorekenaar 'n ander kombinasie 0'e en 1'e gebruik om 'n punt aan te dui nie. Die twee masjiene moet dieselfde binêre taal verstaan.

As die mikrorekenaar en toevoereenheid nie dieselfde taal verstaan nie, kan 'n vertalingstoestel in die rekenaar ingeprop word.

Versyfering van die inligting

'n Mikrorekenaar aanvaar alle soorte inligting. Dit kan egter geen inligting hanteer tensy dit gehelp word om dit in 0'e en 1'e te kodeer nie. Alles moet versyfer (in syfers omskep) word. Dit is betreklik maklik om te begryp hoe dit met 'n toetsbord, waar letters regstreeks gekodeer word, gedoen kan word, maar wat van beelde?

Om 'n tekening in 'n mikrorekenaar te kry, is 'n versyferaar noodsaaklik. Dit lyk soos 'n pen en tekenblad wat met 'n draad aan die mikrorekenaar verbind is. Die "tekenblad" bevat 'n rooster aan die agterkant. Soos in die toetsbord loop drade kruis en dwars onder die oppervlak van die versyferaar en kruis op spesifieke plekke. Wanneer die pen die rooster kruis, registreer die mikrorekenaar die afsonderlike punte waar die roosterdrade kruis.

In sommige gevalle is dit die druk van die pen wat veroorsaak dat 'n punt geregistreer word. In ander gevalle druk 'n mens 'n knop op die elektroniese pen om die pen se posisie te registreer.

Die "muis"

Die "muis" is 'n elektroniese toestel wat in 'n mens se hand gehou word en wat die rooster onder die versyferingsbord aktiveer. Hierdie posisie word dan op die skerm gereflekteer. Dit stel 'n mens in staat om 'n kol, soos met 'n ligpen, maar met die gemak van 'n stuurstok op die skerm rond te beweeg.

▼ Die "muis" besit al die voordele van 'n ligpen, maar is so maklik om te gebruik soos 'n stuurstok. Deur die knoppe op die kas met die hand te druk, kan 'n mens dinge op die skerm rondbeweeg.

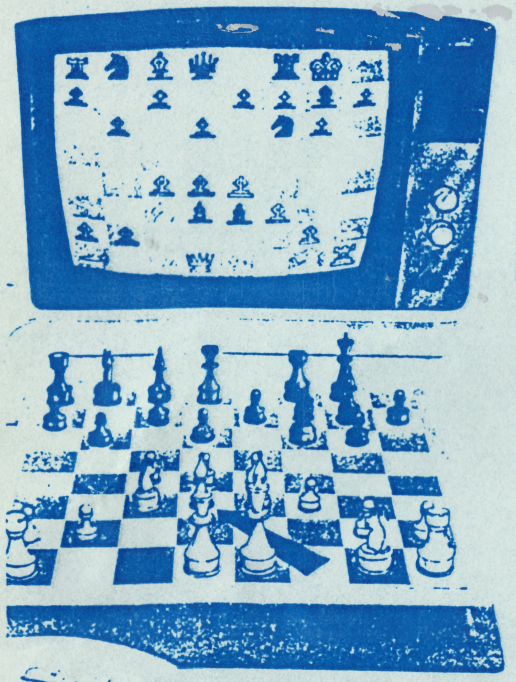
▼ Inligting op baie verpakkings verskyn nou in die vorm van staafkodes. 'n Ligbundel meet die lig wat vanaf die wit strepies tussen die stawe weerkaats word. Die inligting word dan in 'n mikrorekenaar gevoer vir prosessering.

Die "muis"

Now the word processor is upon us, the automatic letter writer spews forth infinities of undesired correspondence, and computer-operated microprinters already inhabiting the planet can turn out 20 000 lines of verbiage a minute, which is 200 times faster than any-

In the beginning was the Word, but must it for ever go on multiplying? Lest someone complain that I am doing my bit, I will subside into silence, and hit the closing key. Full stop.

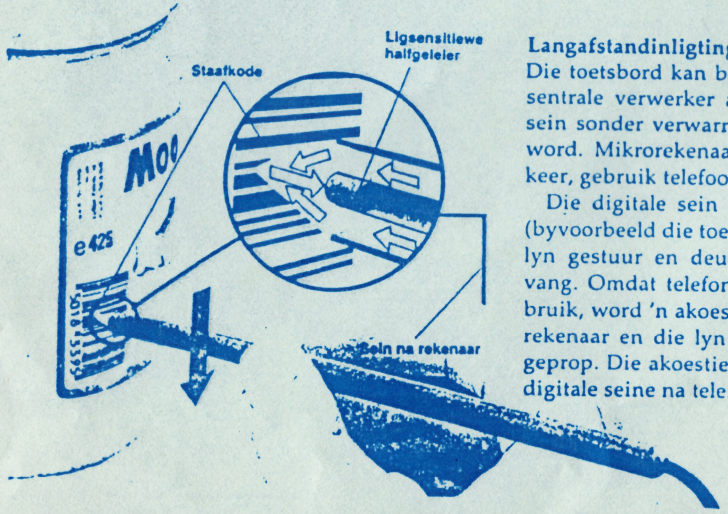
Rekenaarskaak



◄ Baie mikrorekenaars kan skaakposisies op 'n bord herken. As hulle reg geprogrammeer is, kan hulle jou klop. Die geheue hou 'n rekord van jou skuiwe.

Kommunikasie

▲ Mikrorekenaars kan gebruik word om via die telefoon te kommunikeer. Akoestiese koppelaars verander digitale seine in telefoonseine, en omgekeerd.



Langafstandinligting

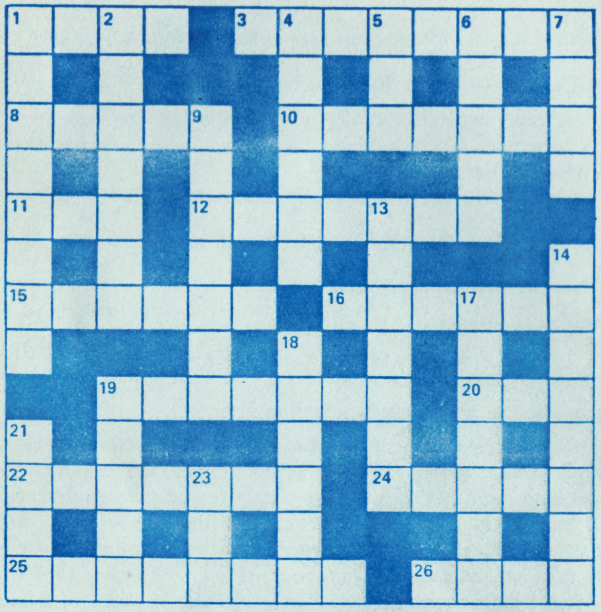
Die toetsbord kan baie kilometers ver van die sentrale verwerker af wees, mits die digitale sein sonder verwarring daarheen gestuur kan word. Mikrorekenaars wat so kan kommunikeer, gebruik telefoonlyne hiervoor.

Die digitale sein vanaf die toevoereenheid (byvoorbeeld die toetsbord) word per telefoonlyn gestuur en deur die mikrorekenaar ontvang. Omdat telefone 'n nie-digitale sein gebruik, word 'n akoestiese koppelaar tussen die rekenaar en die lyn aan beide eindpunte ingeprop. Die akoestiese koppelaar verander die digitale seine na telefoonseine of andersom.

NOTHING is worse than a teacher who knows no more than the students ought to know.

–Goethe

CROSSWORD



ACROSS

1) IF, THEN, , one of BASIC's constructs (4).
2) To send out electromagnetic radiation (8).
8) Cassette access button on a tape recorder (5)
10) Another term for VDU (7).
11) Large manufacturer of magnetic tape (1, 1, 1).
12) A whole number (7).
15) Reuseable memories, usually UV eraseable (6).
16) A power of ten, as in ..... counter (6)
19) Whole, independently moveable graphic objects on a computer screen (7).
20) Frequency measuring instrument (1, 1, 1).
22) Large resistive voltage reducer, as found in older television sets (7).
24) FET terminal (5).
25) Sub-atomic particle (8).
26) Print CHR$(7) and this will sound (4).

DOWN

1) A type of small microphone capsule (8).
2) Audio transducer (7).
4) Input on a tape recorder for distant control of on/off switching (6).
5) One of the two bipolar transistor polarities (1, 1, 1).
6) Electrical value indicator (5).
7) One time round on a coil (4).
9) Variable resistor or capacitor (7)
13) Covered with a film of lubrication (7).
14) Connector, often screwdown, for cables (8).
17) One quarter of 4081 CMOS IC? (3, 4).
18) Two-channel audio (6).
19) Abbreviation for a CRT-based test instrument (5).
21) High-to-low or low-to-high voltage transition (4).
23) Abbreviated name for variable resistor (3).

****
ANSWERS ELSEWHERE IN THIS NEWSLETTER
****

VOLTAGE REGULATORS

The Constant Current — Zener Reference

The effect of zener impedance can be minimized by driving the zener diode with a constant current as shown in Figure 1-4. The value of the zener current is largely independent of Vcc and is given by:

$$I_Z = \frac{V_{BEQ1}}{R_{SC}} \tag{2}$$

where V_{BEQ1} = base-emitter voltage of Q1

This gives a reference voltage of:

$$V_{REF} = V_Z + V_{BEQ1} = V_{BZ} + I_Z Z_Z + V_{BEQ1} \tag{3}$$

where I_Z is constant and given by equation 2.

The reference voltage (about 7 V) of this configuration is therefore largely independent of supply voltage variations. This configuration has the additional benefit of better temperature stability than that of a simple resistor-zener reference.

Referring back to Figure 1-3a, it can be seen that the reference voltage temperature stability is equal to that of the zener diode, VR1. The stability of zener diodes used in most integrated circuitry is about +2.2 mV/°C or ≈ .04%/°C(for a 6.2 V zener). If the junction temperature varies 100°C, the zener, or reference, voltage would vary 4%. A variation this large is usually unacceptable.

However, the circuit of Figure 1-4 does not have this drawback. Here the positive 2.2 mV/°C temperature coefficient (TC) of the zener diode is offset by the negative 2.2 mV/°C TC of the V_{BE} of Q1. This results in a reference voltage with very stable temperature characteristics.

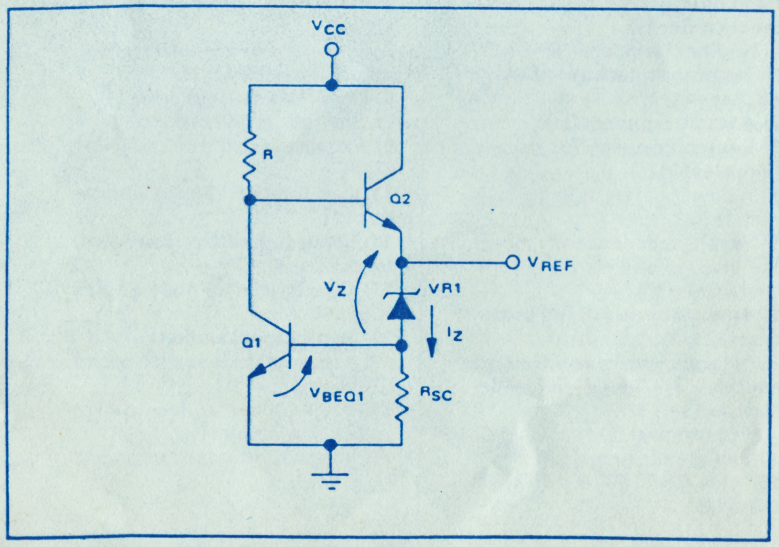


Figure 1-4. Constant Current – Zener Reference

The Bandgap Reference

Although very stable, the circuit of Figure 1-4 does have a disadvantage in that it requires a supply voltage of 9 volts or more. Another type of stable reference which requires only a few volts to operate was described by Widlar[1] and is shown in Figure 1-5. In this circuit V_{REF} is given by:

$$V_{REF} = V_{BEQ3} + I_2 R_2 \tag{4}$$

where $I_2 = \frac{V_{BEQ1} - V_{BEQ2}}{R_1}$ (neglecting base currents)

The change in V_{REF} with junction temperature is given by:

$$\Delta V_{REF} = \Delta V_{BE3} + \left\{\frac{\Delta V_{BEQ1} - \Delta V_{BEQ2}}{R_1}\right\} R_2 \tag{5}$$

It can be shown that,

$$\Delta V_{BEQ1} = \Delta T_j \, K \, \ln I_1 \tag{6}$$

and $$\Delta V_{BEQ2} = \Delta T_j \, K \, \ln I_2 \tag{7}$$

where K = a constant

ΔT_j = change in junction temperature

and $I_1 > I_2$

Combining (5), (6), and (7)

$$\Delta V_{REF} = \Delta V_{BEQ3} + \Delta T_j K \left(\frac{R_2}{R_1}\right) \ln \frac{I_1}{I_2} \tag{8}$$

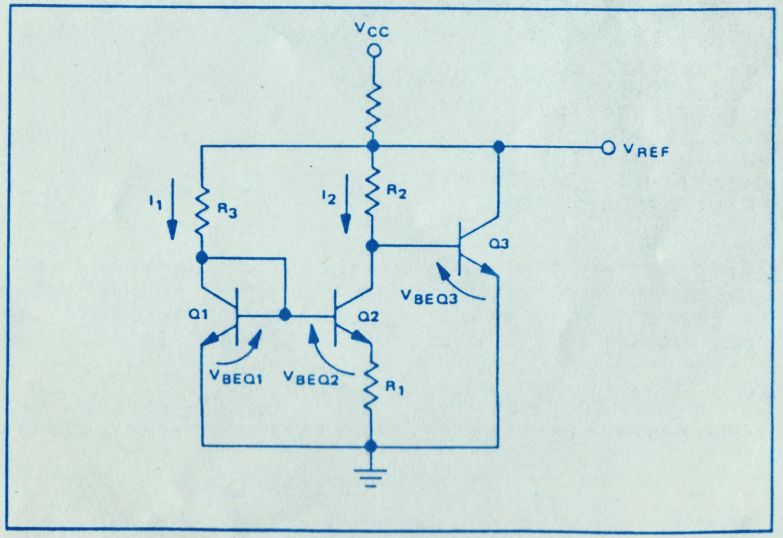


Figure 1-5. Bandgap Reference

Since ΔV_{BEQ3} is negative, and with $I_1 > I_2$, $\ln I_1/I_2$ is positive, the net change in V_{REF} with temperature variations can be made to equal zero by appropriately selecting the values of I_1, R_1, and R_2.

```
**********************************************************
*                                                        *
*ROUNDING OFF NUMBERS AND GETTING THE CORRECT ANSWER     *
*                                                        *
**********************************************************

WHEN WORKING WITH RANDS AND CENTS, IT IS OFTEN FOUND
THAT THE ROUNDING OFF OF NUMBERS USING ###.## FORMAT
(OR SIMILAR), PRODUCES THE LOSS OR GAIN OF ONE CENT,
IN SUBSEQUENT CALCULATIONS.
THIS IS DUE TO THE FACT THAT A NUMBER IS RETAINED IN
ITS ORIGINALLY ENTERED FORM, REGARDLESS OF ANY PRINT
USING FORMATS FOR DISPLAY.
ONE OF THE WAYS TO OVERCOME THIS PROBLEM IS TO  WORK
IN CENTS (WHOLE NUMBERS) AND THEREFORE ALLOW THE USE
OF THE <INTEGER> FUNCTION.
PRINTING OUT OF VALUES IN RANDS AND CENTS IS DONE BY
DIVIDING THE CENTS BY 100, WHEREAS ALL  CALCULATIONS
WILL RETAIN THEIR ACCURACY IN SPITE OF <ROUNDING> OF
NUMBERS BY MEANS OF THE PRINTUSING FORMATTING.

THIS LITTLE PROGRAM WILL ILLUSTRATE THIS PRINCIPLE.
AS WITH ALL <INTEGER> CALCULATIONS, REMEMBER TO ADD
0.5 SO AS TO GET A WHOLE NUMBER WHICH IS GREATER IF
THE LAST NUMBER IS 5 OR LARGER. ALTERNATIVELY,  THE
LESSER WHOLE NUMBER IF THE LAST DIGIT IS 4 OR LESS.


10 A$="###.##"
20 B=19.855*100
30 PRINTUSINGA$:B/100
40 C=10.154*100
50 PRINTUSINGA$:C/100
60 D=INT(B+.5)-INT(C+.5)
70 PRINTUSINGA$;D/100


NOTE THAT WITHOUT USING THE <INTEGER> FUNCTION, THE
ANSWER WOULD HAVE BEEN   9.70   WHICH ALTHOUGH BEING
QUITE CORRECT IN NORMAL CALCULATION, IS INCORRECT IF
YOU ARE WORKING IN ROUNDED-OFF RANDS AND CENTS.

TA-TA

**********************************************************
```

WE APPRECIATE frankness from those who like us. Frankness from others is called insolence.

—Andre Maurois

Solution to Crossword

Across:

1) ELSE
3) Transmit
8) Eject
10) Monitor
11) TDK
12) Integer
15) EPROMs
16) Decade
19) Sprites
20) DFM
22) Dropper
24) Drain
25) Electron
26) Bell

Down:

1) Electret
2) Speaker
4) Remote
5) NPN
6) Meter
7) Turn
9) Trimmer
13) Greased
14) Terminal
17) AND gate
18) stereo
19) 'Scope
21) Edge
23) Pot

NUMBER ROUNDER

Digital computers are great. They perform thousands of computations in almost no time at all and turn out results that are accurate to many decimal places. However, most of the time this accuracy is not necessary and, in fact, can be bothersome. If you're producing a list of numbers by computer and some have eight decimal places while others have only two, it really messes up the appearance of the number columns in your final output. To overcome those super-long numbers, it is convenient to round off.

This program will allow you to input any number and round off to any fewer number of decimal places (within the limits of your computer, of course). Most computers will not handle more than eight places after the decimal point. So if you enter a number with eleven places after the decimal point and try to round off to nine decimal places, don't be surprised when you find that your answer is rounded off to eight places. This is because the computer has automatically rounded off to the limit of its accuracy. There are, however, some BASIC interpreters that just truncate the extra decimal places. So pay attention to your computer's capabilities.

```
1  PRINT : PRINT : PRINT : PRINT "WRITTEN BY JULES H. GILDER"
2  PRINT : PRINT
3  FOR X = 1 TO 4000: NEXT X
10  PRINT : PRINT : PRINT : PRINT
20  PRINT  TAB( 13)"NUMBER ROUNDER"
30  PRINT : PRINT : PRINT : PRINT
40  PRINT "THIS PROGRAM WILL ROUND OFF ANY GIVEN"
50  PRINT "NUMBER TO ANY NUMBER OF PLACES."
60  PRINT : PRINT : PRINT
70  INPUT "ENTER NUMBER  ";N
75  PRINT
80  INPUT "ROUND TO HOW MANY-PLACES  ";R
90  PRINT : PRINT : PRINT
95 RR =  INT (10 ↑ R + .5)
96  IF  R = 6  THEN  RR = 1E6
98 N2 =  INT (RR * N + .5)
100 N1 = N2 / RR
110  PRINT N;" ROUNDED TO ";R;" PLACES IS"
120  PRINT : PRINT N1
130  END


ENTER NUMBER  34.3086516
ROUND TO HOW MANY PLACES  4

34.3086516 ROUNDED TO 4 PLACES IS
34.3087
```

Trump Card

MY FATHER asked my mother what she wanted as a birthday gift. "Anything," she said with a smile, "as long as it has diamonds in it."

On her birthday, my mother opened her present. It was a pack of cards.

—Silvia Boaga, Durban